Symmetric Encryption Schemes: DES

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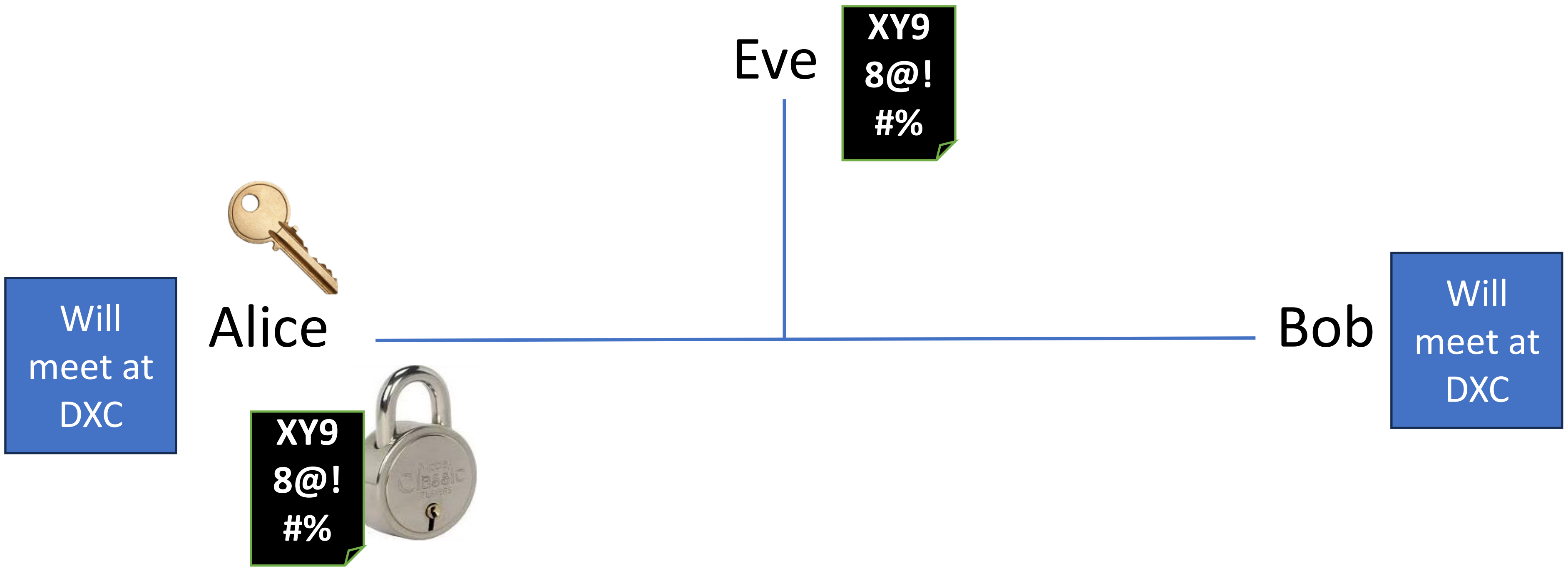
Symmetric key cryptography



Cryptanalysis

- **Brute force:** Try every key
- **Ciphertext-only attack:**
  - Attacker knows the ciphertext of several messages encrypted with the same key.
  - Possible to recover plaintext (also possible to deduce the key) by looking at the frequency of letters.
- **Ciphertext Only attack on Caesar Cipher:**
  - Ciphertext: ZHOFRPH WR WKH FLSKHU FODVV
  - Most common trigram in English: THE
  - Digrams: TO, IS, IN, HE
  - Attacker gets the key!!!

Cryptanalysis Example

- **Known-plaintext attack:**
  - Attacker observes pairs of plaintext/ciphertext encrypted with the same key.
  - Possible to deduce the key and/or devise algorithm to decrypt the ciphertext.

- **Known plaintext attack on XOR_Algo:**
  - Suppose attacker gets a pair (CT: 0110, PT:1010)
  - Key: **1100**

Cryptanalysis

- **Chosen-Plaintext attack:**
  - Attacker can choose the plaintext and look at the paired ciphertext.
  - Attacker has more control than known-plaintext attack and maybe able to gain more information about the key.
- **Adaptive Chosen-Plaintext attack:**
  - Attacker chooses a series of plaintexts, basing the next plaintext on the result of the previous encryption.
- **Chosen-Ciphertext attack:**
  - Attacker can choose ciphertexts and get the corresponding plaintext from a decryption oracle.
  - The attacker can then use this information to try to recover the secret key used for decryption.
    - Caesar cipher:
      - Query with ciphertext: GGGG
      - suppose the PT output is JJJJ, then key will be 3

Cryptanalysis – Chosen Plain-Text Attack

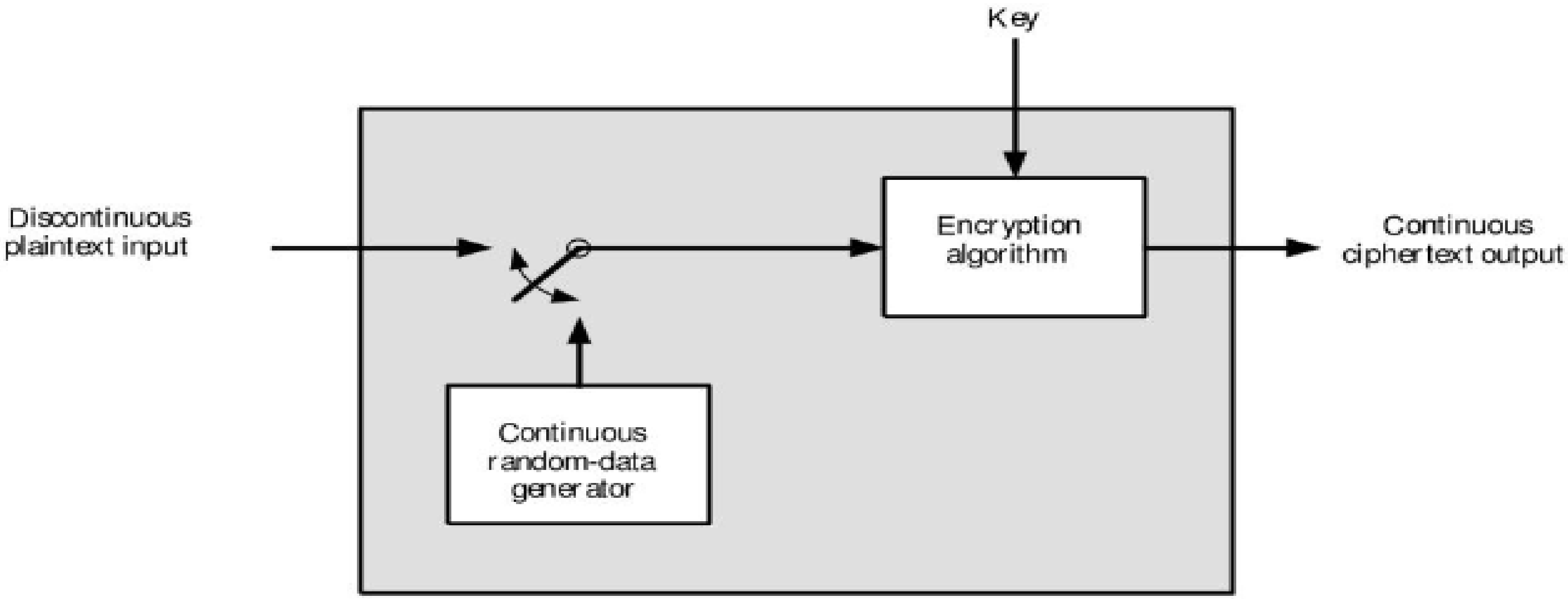
```
Input : Plaintext :   GEEKSFORGEEKS
             Keyword :  AYUSH
Output : Ciphertext :  GCYCZFMLYLEIM
```

- The attacker chooses a plaintext : AAAAAAAAAA (10 characters)
- Let's say the ciphertext is: AYUSHAYUSHAY
- Derive the key: AYUSH

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

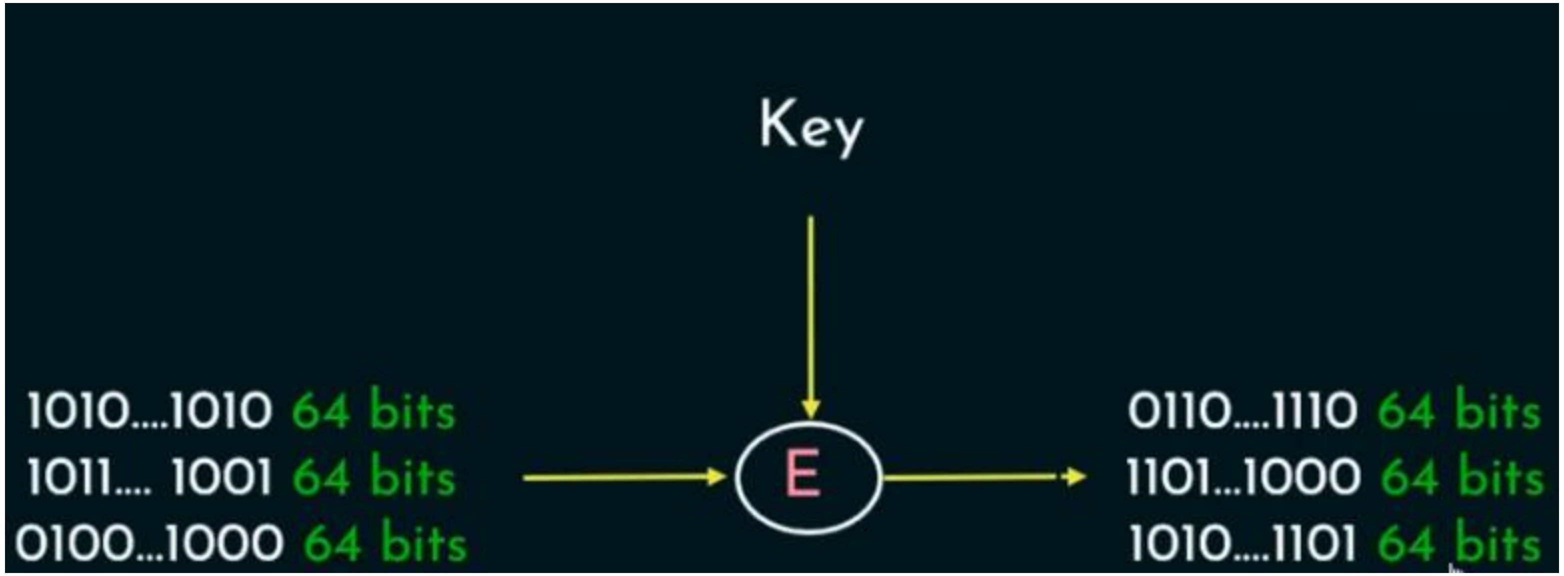
Traffic Padding

- Traffic Padding is a technique used to add extra data to a network traffic stream in an attempt to obscure the true content of the traffic and make it more difficult to analyze.



Block Cipher

- A block cipher encrypts a **block of given plaintext** using a **key** and a **cryptographic algorithm**.
- A block cipher processes the data blocks of fixed size (64 bit, 128 bit, etc.).
- Typically, a message's size exceeds a block's size.
- As a result, the lengthy message is broken up into a number of sequential message blocks, and the cipher operates on these blocks one at a time.



Plain Text Encoding

- Each character in the plaintext "Hello Mike" is represented by a single byte in the ASCII encoding, where 1 byte equals 8 bits.

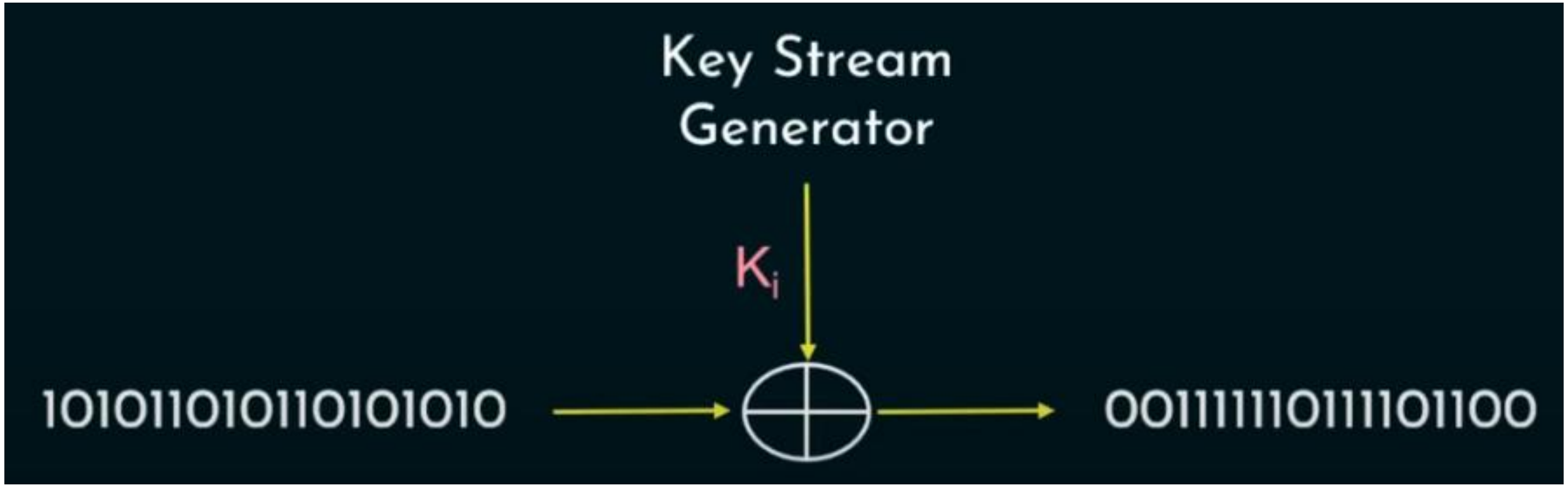
Character	ASCII (Decimal)	Binary	Hexadecimal
H	72	01001000	48
e	101	01100101	65
l	108	01101100	6C
l	108	01101100	6C
o	111	01101111	6F
(space)	32	00100000	20
M	77	01001101	4D
i	105	01101001	69
k	107	01101011	6B
e	101	01100101	65

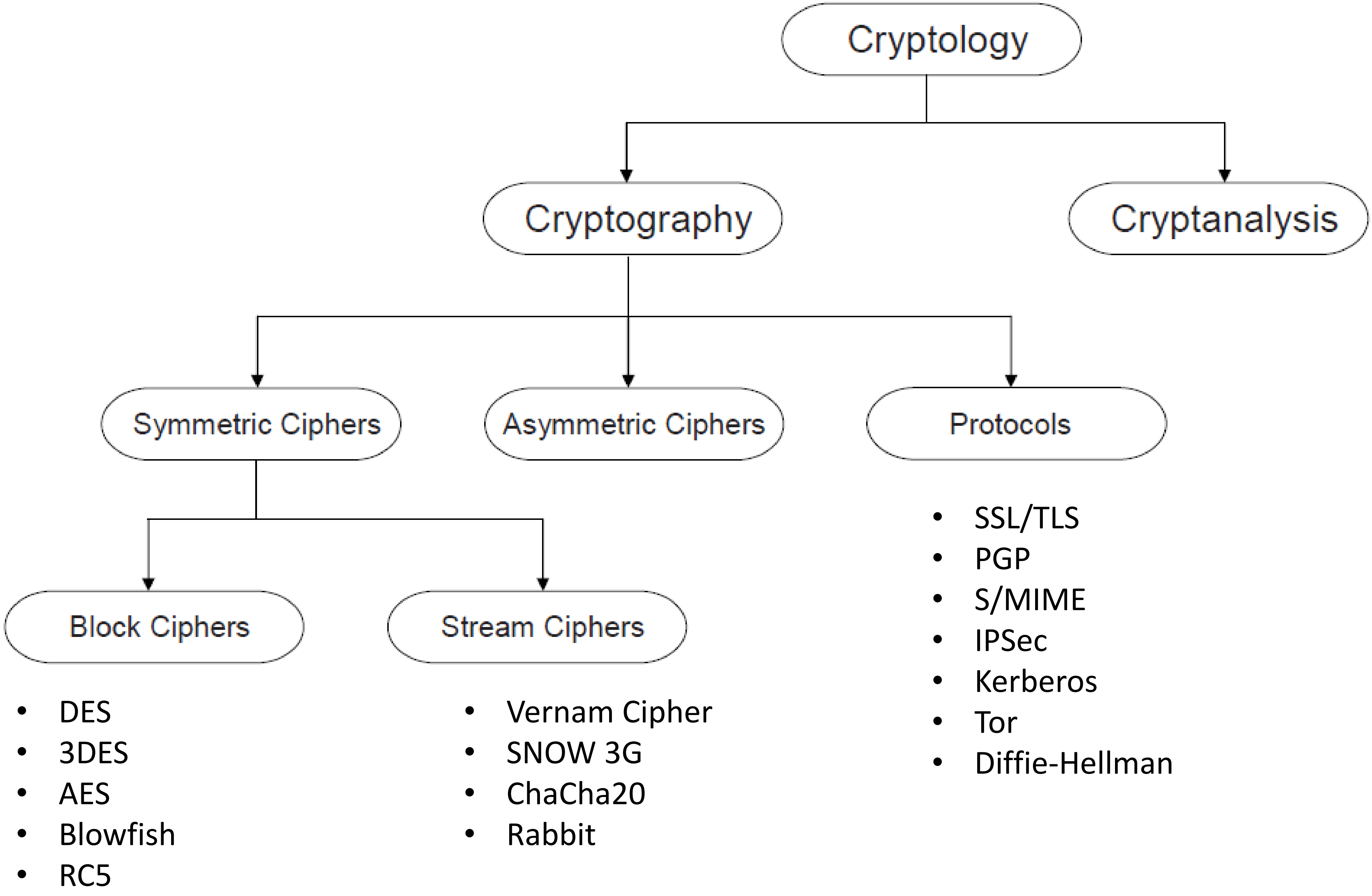
Padding of Plain text

- Block Size: DES operates on blocks of 64 bits (which is 8 bytes).
- This means that the input plaintext must be in multiples of 8 bytes.
  - Example:
  - Plaintext: "Hello Mike" (10 characters)
  - Length: 10 bytes (since each character is 1 byte).
- Padding Required: 16 bytes (next multiple of 8) - 10 bytes (original length) = 6 bytes of padding.
  - Padding Value: In PKCS#7 padding, if 6 bytes are added, each byte will be 0x06 in hexadecimal.
  - Original Plaintext: "Hello Mike" (10 bytes)
  - Padded Plaintext: "Hello Mike\x06\x06\x06\x06\x06\x06" (16 bytes)

Stream Cipher

- A stream cipher is a cryptographic algorithm that encrypts messages in a continuous stream, one bit of data at a time.
- It encrypts a given plain text to a cipher text using a secret key.
- It can sometimes do the encryption in a continuous stream, by taking one byte of data at a time.
- E.g:- Vernam Cipher



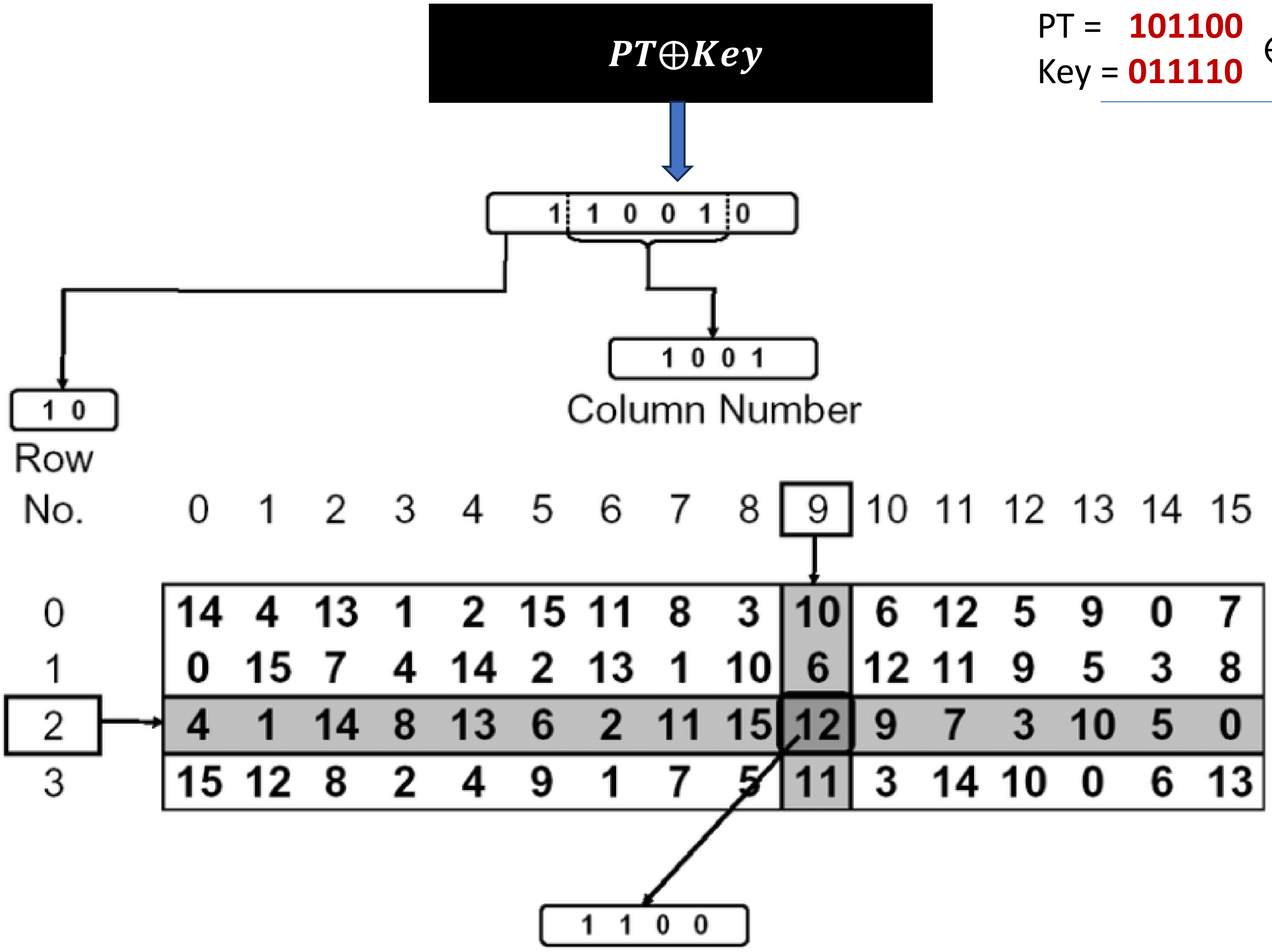

Cryptology
Cryptography
Cryptanalysis
Symmetric Ciphers
Asymmetric Ciphers
Protocols
Block Ciphers
Stream Ciphers
• DES
• 3DES
• AES
• Blowfish
• RC5
• Vernam Cipher
• SNOW 3G
• ChaCha20
• Rabbit
• SSL/TLS
• PGP
• S/MIME
• IPSec
• Kerberos
• Tor
• Diffie-Hellman

Confusion

- **Caesar cipher:**
  - Given ciphertext only, we can derive the key based on statistical analysis.
  - There is a linear relationship between the ciphertext and the key.
- Confusion means that **each binary digit (bit) of the ciphertext should depend on several parts of the key**, obscuring the connections between the two.
- The property of confusion hides the relationship between the ciphertext and the key.
- Confusion maybe attained using Substitution boxes (S-boxes).

S-Box Example

PT = **101100**
Key = **011110** ⊕



Row No.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S-box and Confusion

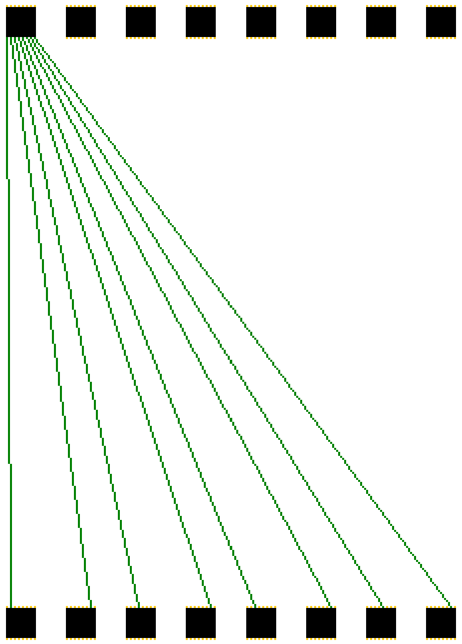
- S-boxes are small lookup tables that replace input bits with output bits in a non-linear and unpredictable way.
- **Non-linearity:**
  - This non-linearity ensures that even if an attacker knows part of the ciphertext or key, they can't easily predict how the other parts related.
- **Multiple Dependencies:**
  - Each output bit from the S-box depends on several bits of the input, making the relationship between input (key) and output (ciphertext) highly complex.

Diffusion

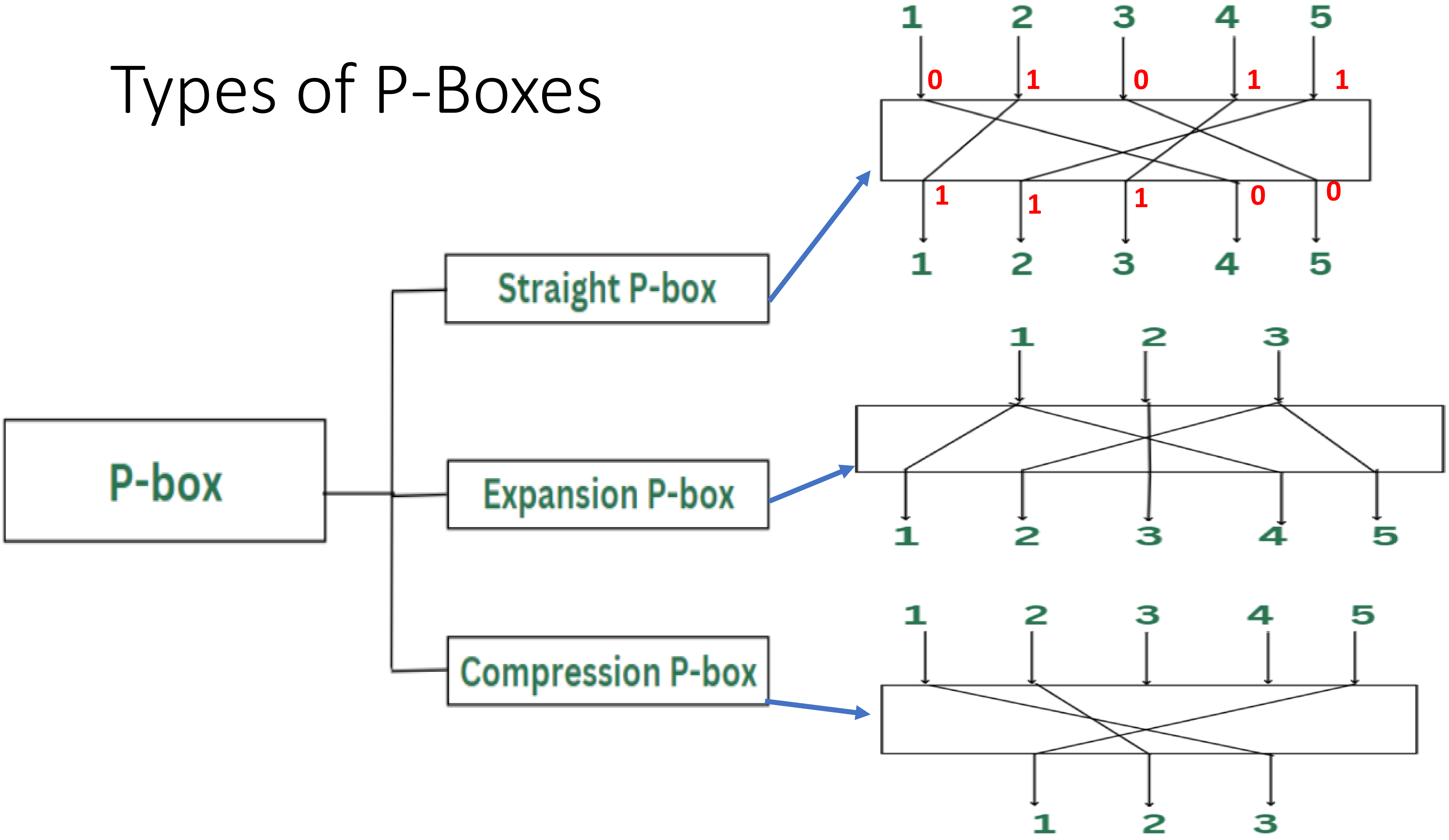
- In Caesar cipher, suppose the:
  - Plain Text = **THE** CAT SLEPT ON **THE** MAT.
  - Cipher Text = **WKH** FDW VOHSW RQ **WKH** PDW
  - **Patterns in the plaintext are visible in the ciphertext.**
- **Diffusion** refers to dissipating the statistical structure of plaintext over the bulk of ciphertext.
- For example, **diffusion ensures that any patterns in the plaintext, such as redundant bits, are not apparent in the ciphertext**.
- Diffusion is provided by permutation boxes.
- If we change **a single bit of the plaintext**, then **about half of the bits in the ciphertext should change**, and vice versa.

Permutation-Box (P Box)

- An example of a 64-bit P-box which spreads the input S-boxes to as many output S-boxes as possible.
- 1100 0101 1101 0000
- 1010 1110 0000 0110

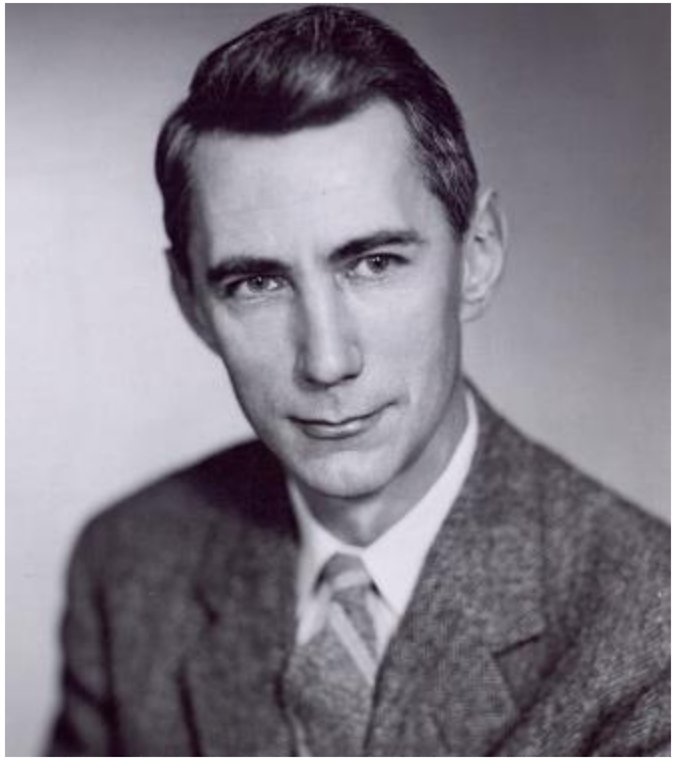


Types of P-Boxes



Claude Shannon

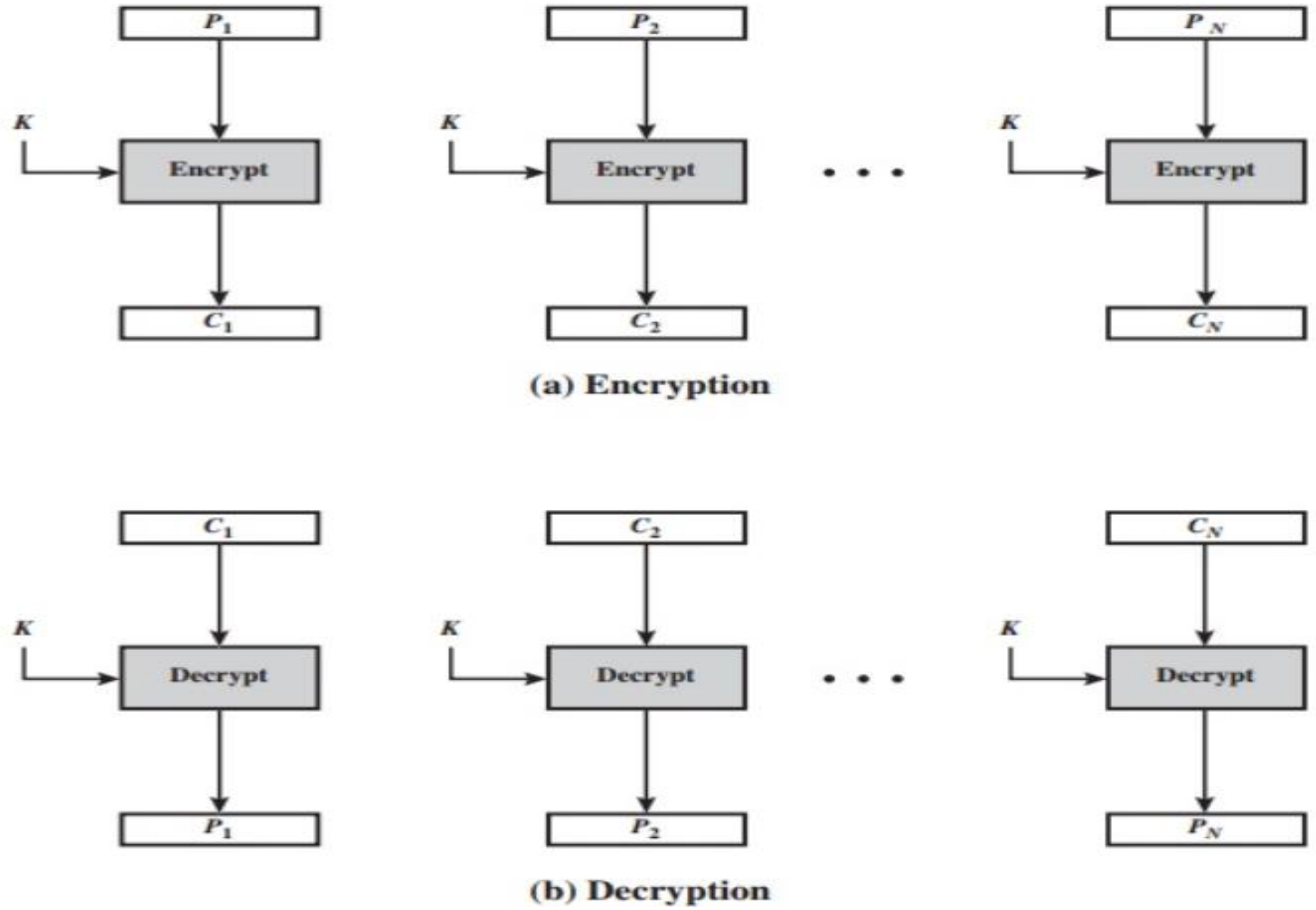
- Father of information theory.
- Proposed the idea of **confusion** and **diffusion**.
- Year: 1945
- Complexity is implemented through a **well-defined** and **repeatable series** of **substitutions and permutations**.
- Multiple repeated rounds of S-Box and P-Box.



Block Cipher Modes of Operation

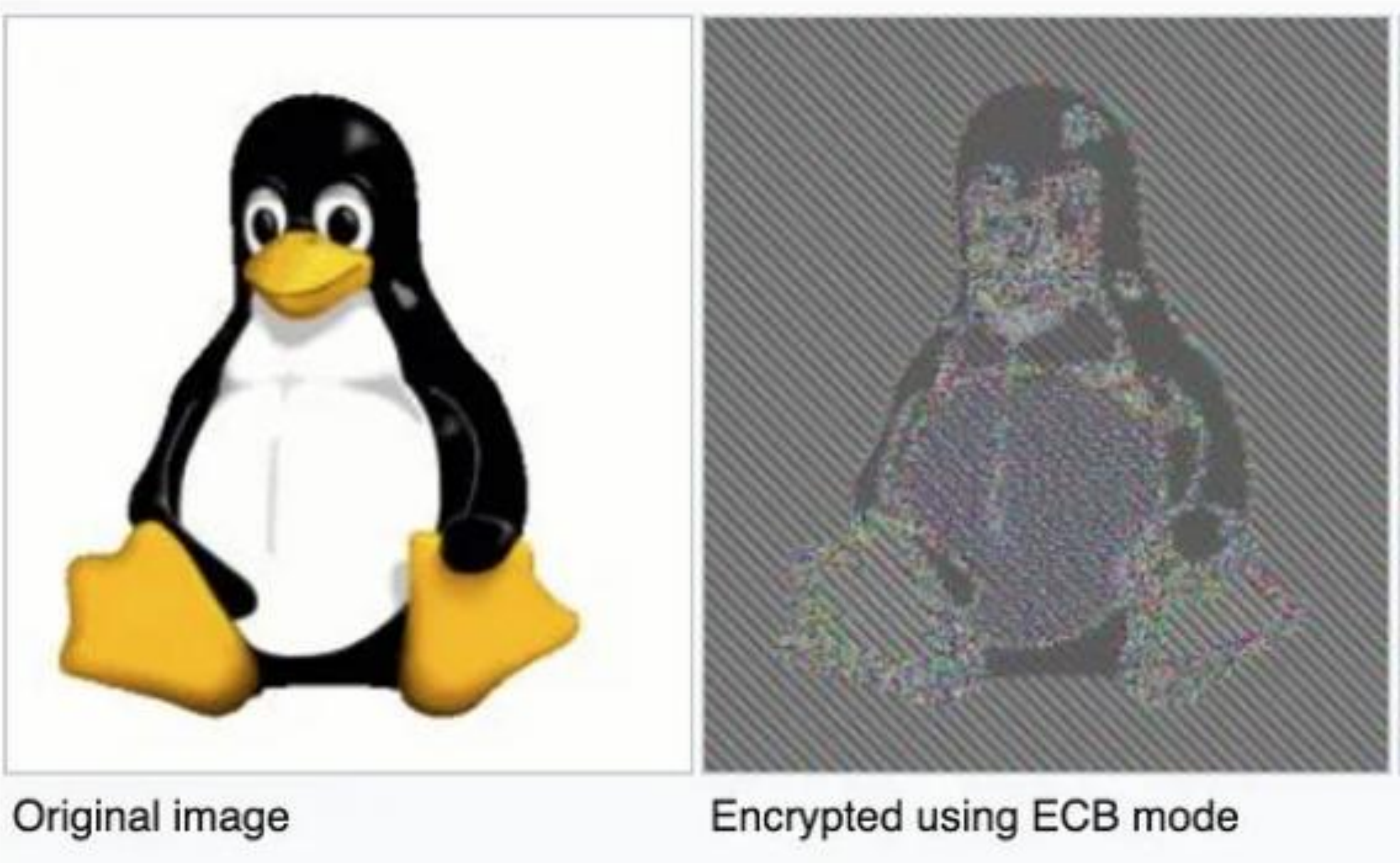
- A block cipher operates on fixed-size blocks (e.g., 64 bits in DES, 128 bits in AES).
- But most real-world data is longer or shorter than a single block.
- Modes of operation allow block ciphers to handle data of any length by processing multiple blocks or padding the last block if necessary.
- Types:
  - Electronic Code Book
  - Cipher Feedback
  - Cipher Block Chaining
  - Output Feedback
  - Counter

Electronic Code Book (ECB)



- Plaintext is handled one block at a time and each block of plaintext is encrypted using the same key.
- The most significant characteristic of ECB is that if the same b-bit block of plaintext appears more than once in the message, it always produces the same ciphertext.
- For lengthy messages, the ECB mode may not be secure.

ECB-Penguin



- Identical plaintext blocks produce identical ciphertext blocks, which can leak patterns and make it less secure for many applications.

CrypTool 1.4.42 - startingexample-en.txt

File Edit View Encrypt/Decrypt Digital Signatures/PKI

startingexample-en.txt

BAT BAT BAT BAT

Encrypted with DES ECB

DES (ECB) encryption of <startingexample-en.txt>, key <12 12 12 12 12 12 12 12>

```
00000000   F1 95 42 D0 85 7A CF 56 F1 95 42 D0 85 7A CF 56 DF 69 AD E7 19 1E A0 01      ..B..z.V..B..z.V.i......
```

Encrypted with DES CBC

DES (CBC) encryption of <startingexample-en.txt>, key <12 12 12 12 12 12 12 12>

```
00000000   F1 95 42 D0 85 7A CF 56 CE 28 F7 0A 82 3C 7F 83 CC BF 88 19 4F 41 20 C4      ..B..z.V.(...<......OA .
```

startingexam

BAT

DES (CBC) encryption of <startingexample-en.txt>, key <12 12 12 12 12 12 12 12>

```
00000000   52 11 D5 8C 39 98 F8 57      R...9..W
```

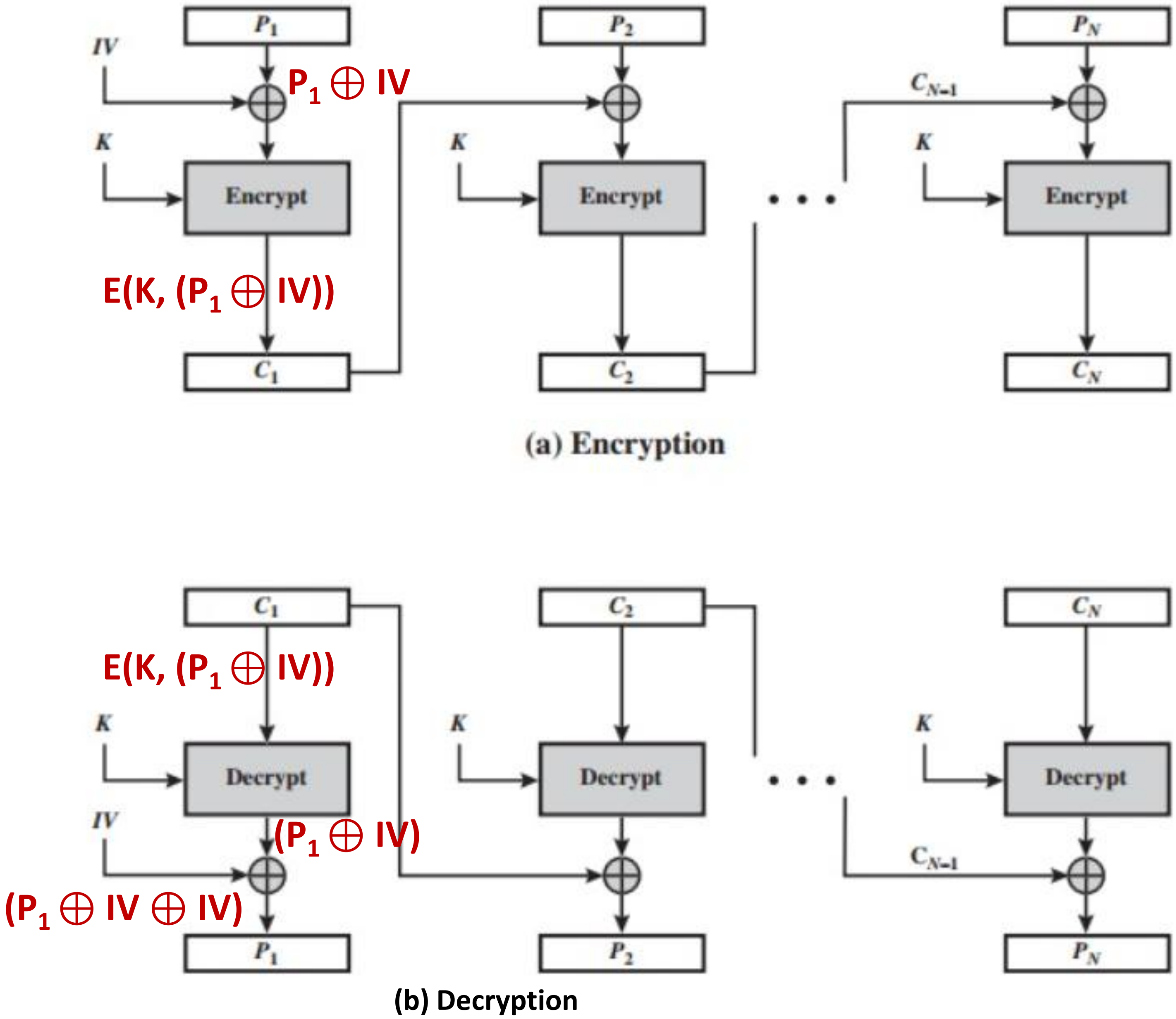
startingexar

CAT

DES (CBC) encryption of <startingexample-en.txt>, key <12 12 12 12 12 12 12 12>

```
00000000   74 5A 72 A4 B8 F4 5B 0E      tZr...[
```

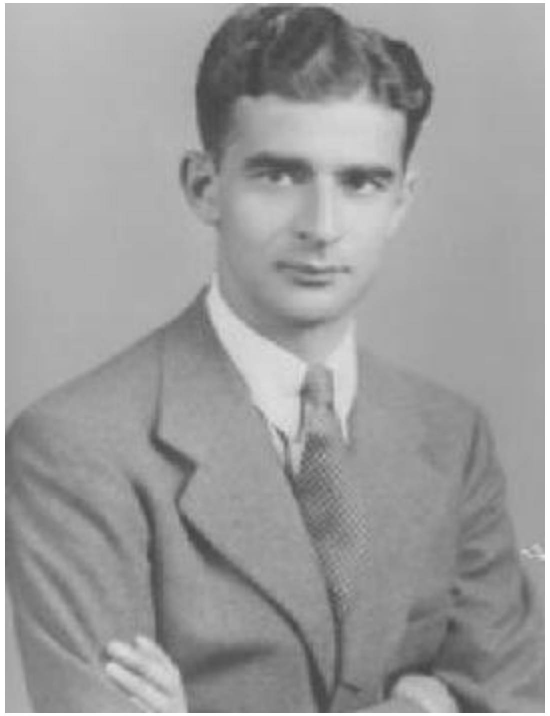
Cipher Block Chaining



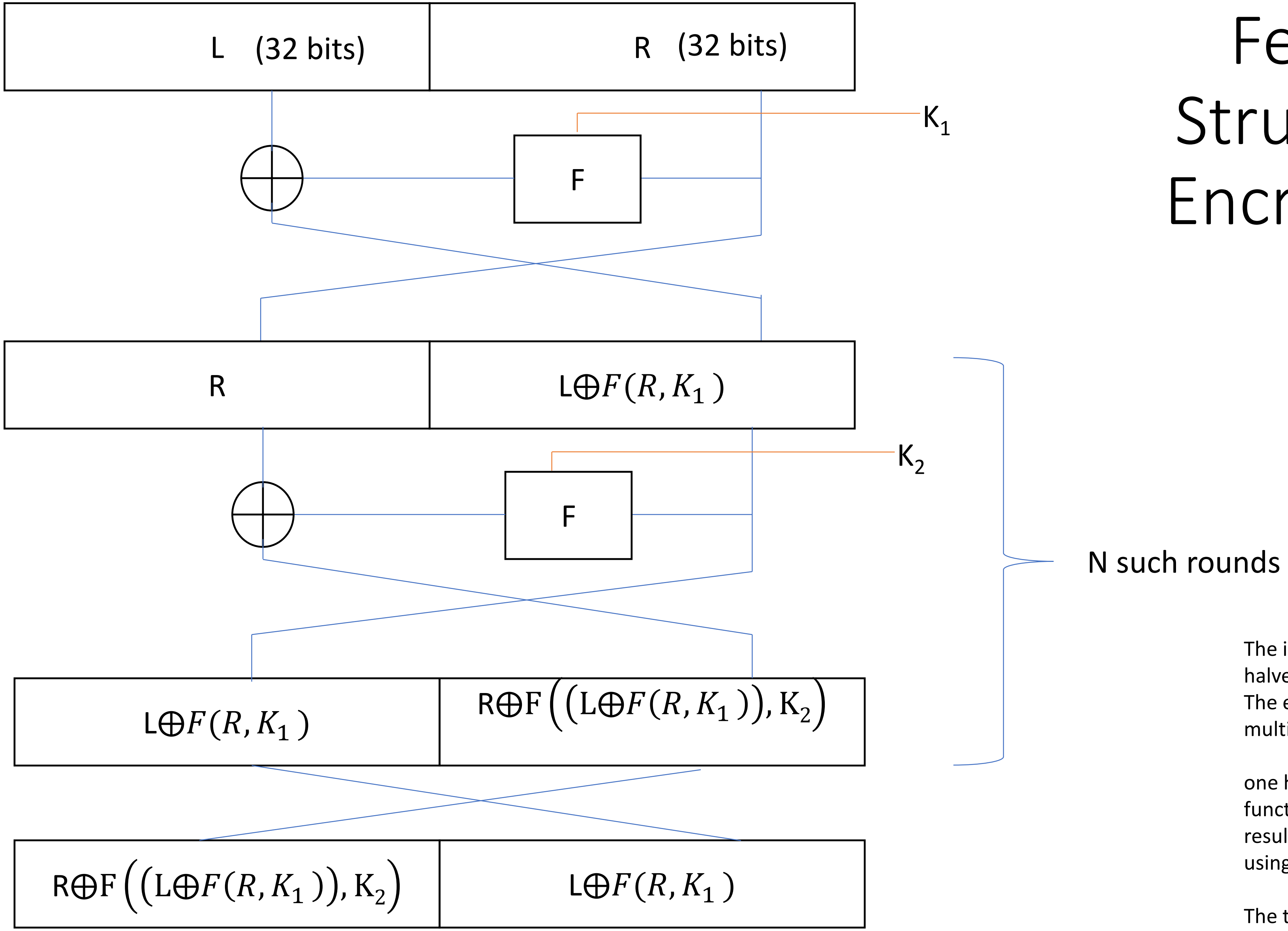
- IV – Initial value of 64 bits
- The input to the encryption algorithm is the XOR of the current plaintext block and the preceding ciphertext block;
- the same key is used for each block.
- Therefore, repeating patterns of b bits are not exposed.
- As with the ECB mode, the CBC mode requires that the last block be padded to a full b bits if it is a partial block.

Feistel Structure (1970)

- **Horst Feistel** : German-American cryptographer who worked on the design of ciphers at IBM.
- A Feistel structure is a **design framework** used in many symmetric encryption algorithms (DES, blowfish, etc.).
  - The input block of data is split into two halves.
  - The encryption process involves multiple rounds, where in each round:
    - one half is transformed using a round function and a subkey, and then the result is combined with the other half using an XOR operation.
  - The two halves are then swapped before the next round.
  - This structure ensures that the decryption process mirrors the encryption process, making it efficient and secure.



Feistel Structure-Encryption

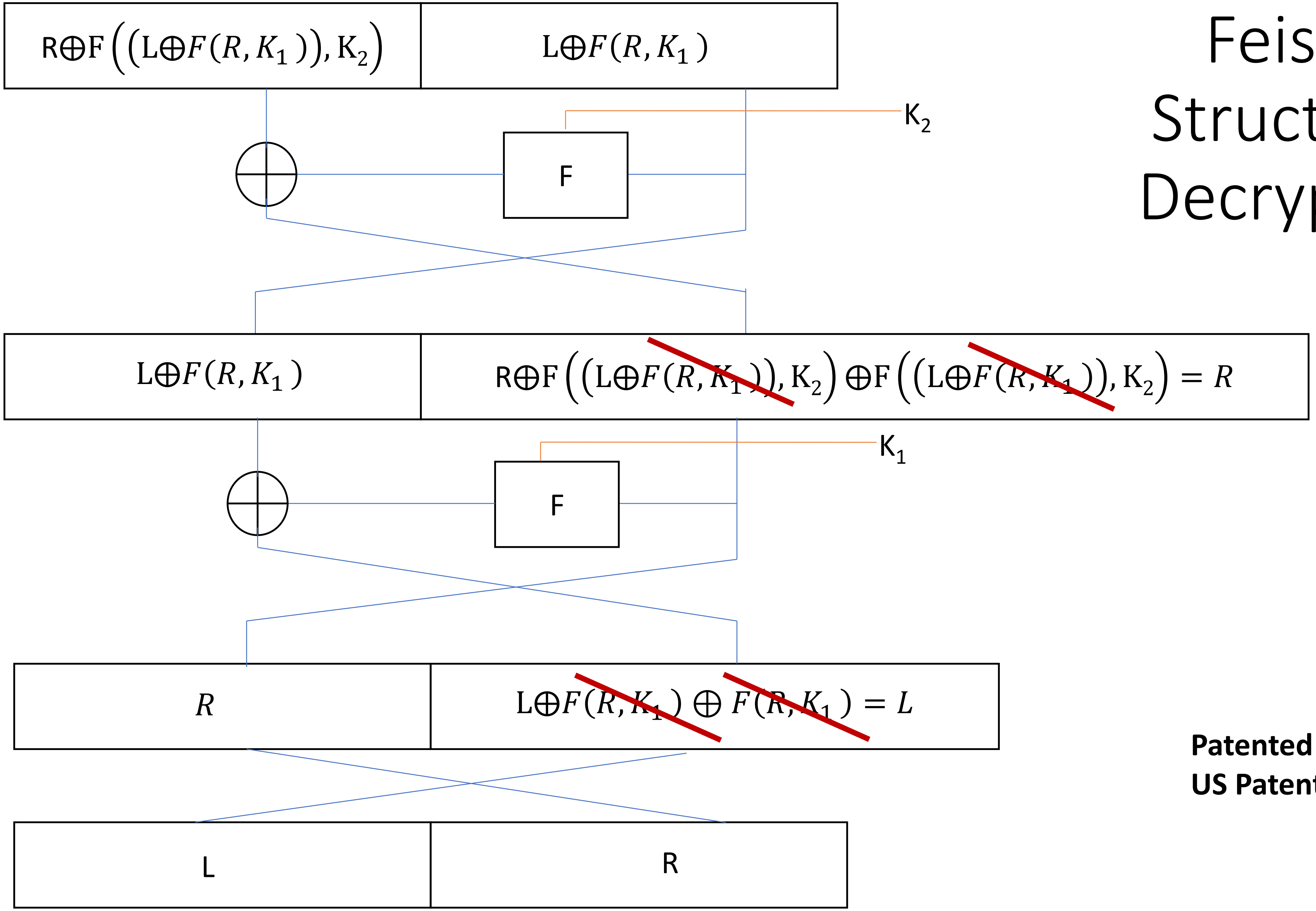


The input block of data is split into two halves.
The encryption process involves multiple rounds, where in each round:

one half is transformed using a round function and a subkey, and then the result is combined with the other half using an XOR operation.

The two halves are then swapped before the next round.

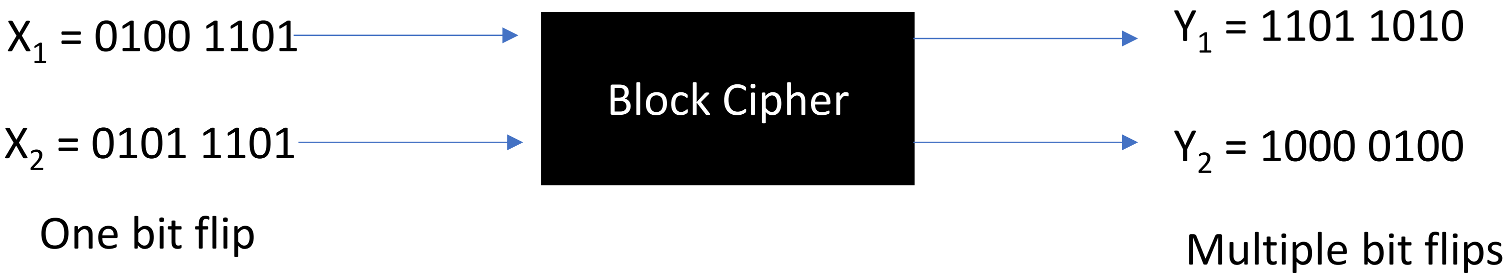
Feistel Structure- Decryption

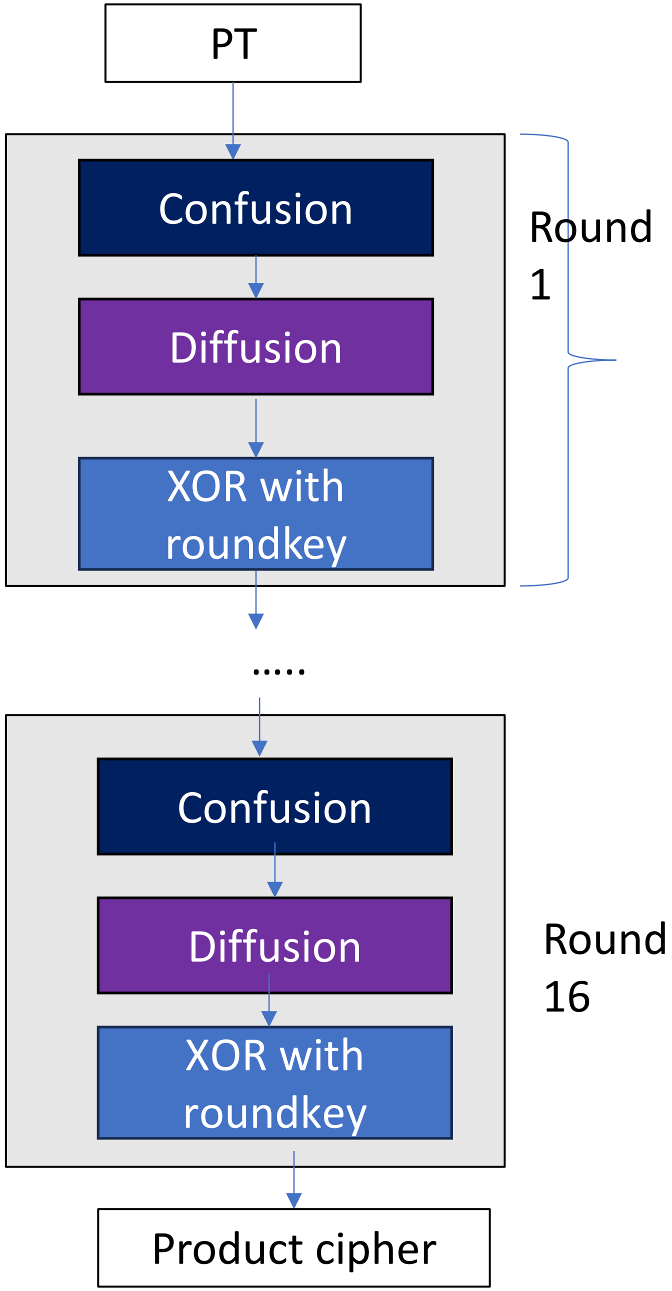


Patented in 1973
US Patent 3798359

DES: Product Cipher

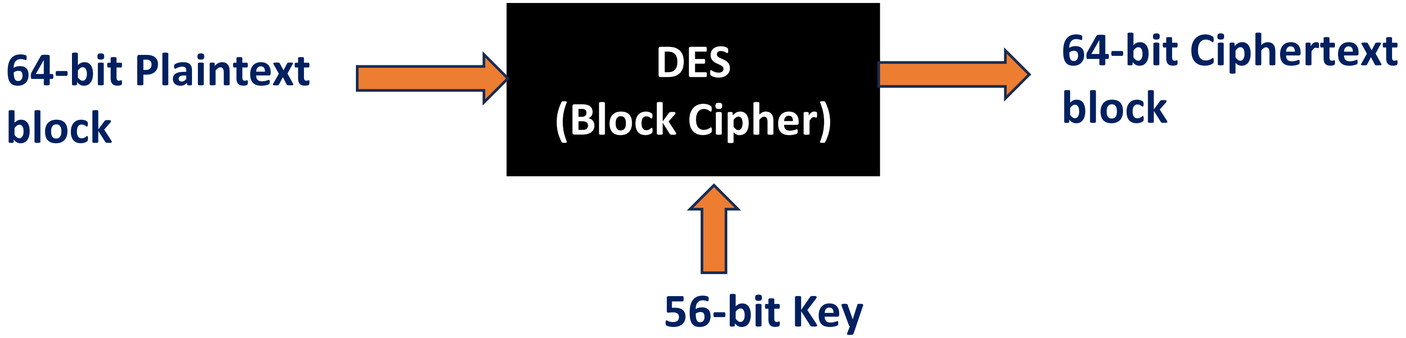
- Most of today's block ciphers are product ciphers as they consist of rounds which are applied repeatedly to the data.
- Can reach excellent diffusion:





Data Encryption Standard

- Symmetric Block Cipher, developed by IBM with input from NSA in 1977.
- Standardized by the U.S. government in 1977 for commercial use.
- Widely trusted through the 1980s and early 1990s.
- Publicly broken (practically) in 1998.
- Implementation of Feistel Structure.



Data Encryption Standard

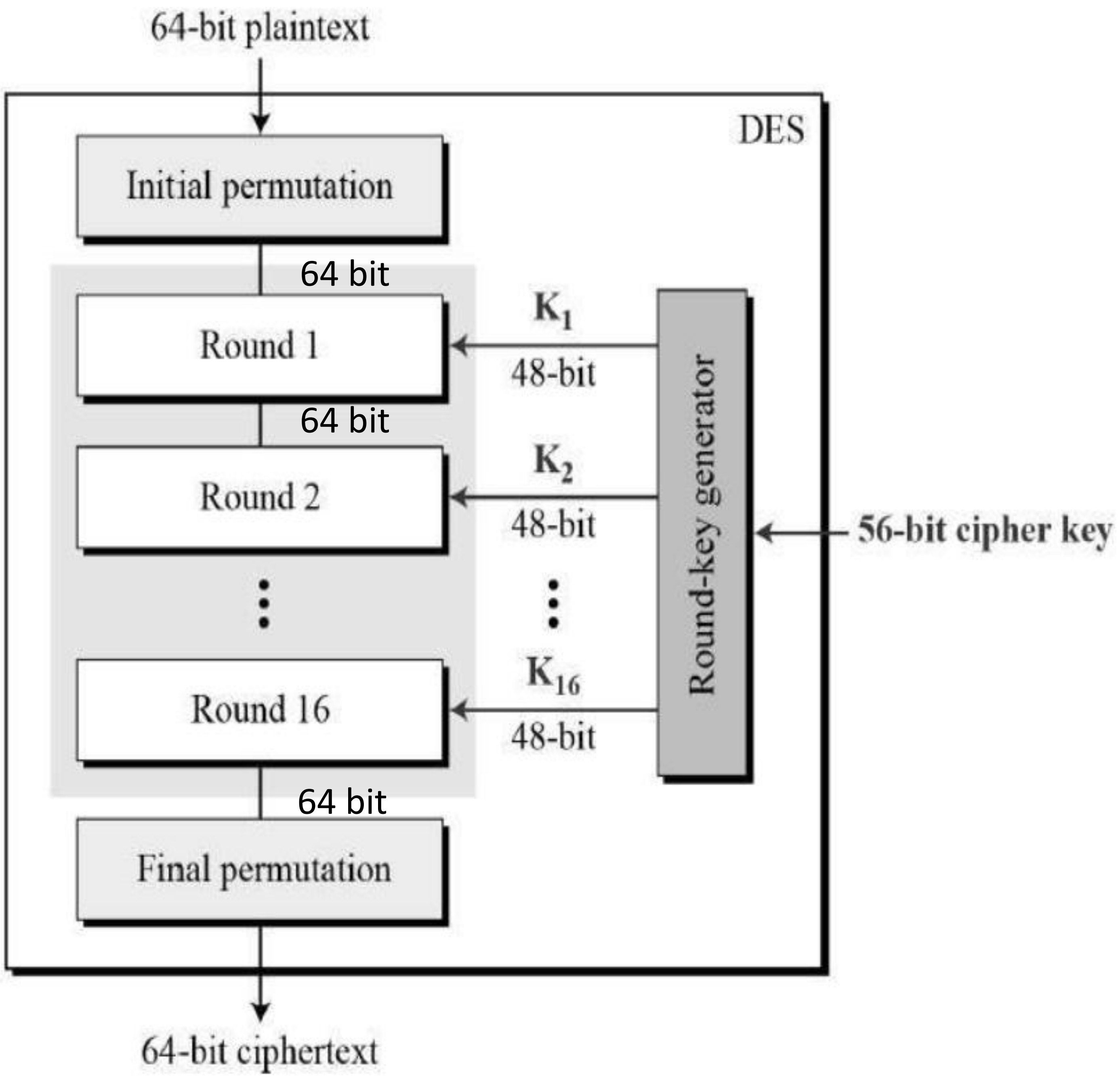
- Plaintext block size = 64 bits.
- Ciphertext block size = 64 bits.
- Main key = 64 bits
- Sub key = 56 bits
- Size of the round key = 48 bits
- No: of rounds = 16 rounds.
- DES is replaced by the Advanced Encryption Standard in 2001.
- DES is a product cipher.

Putting it all together

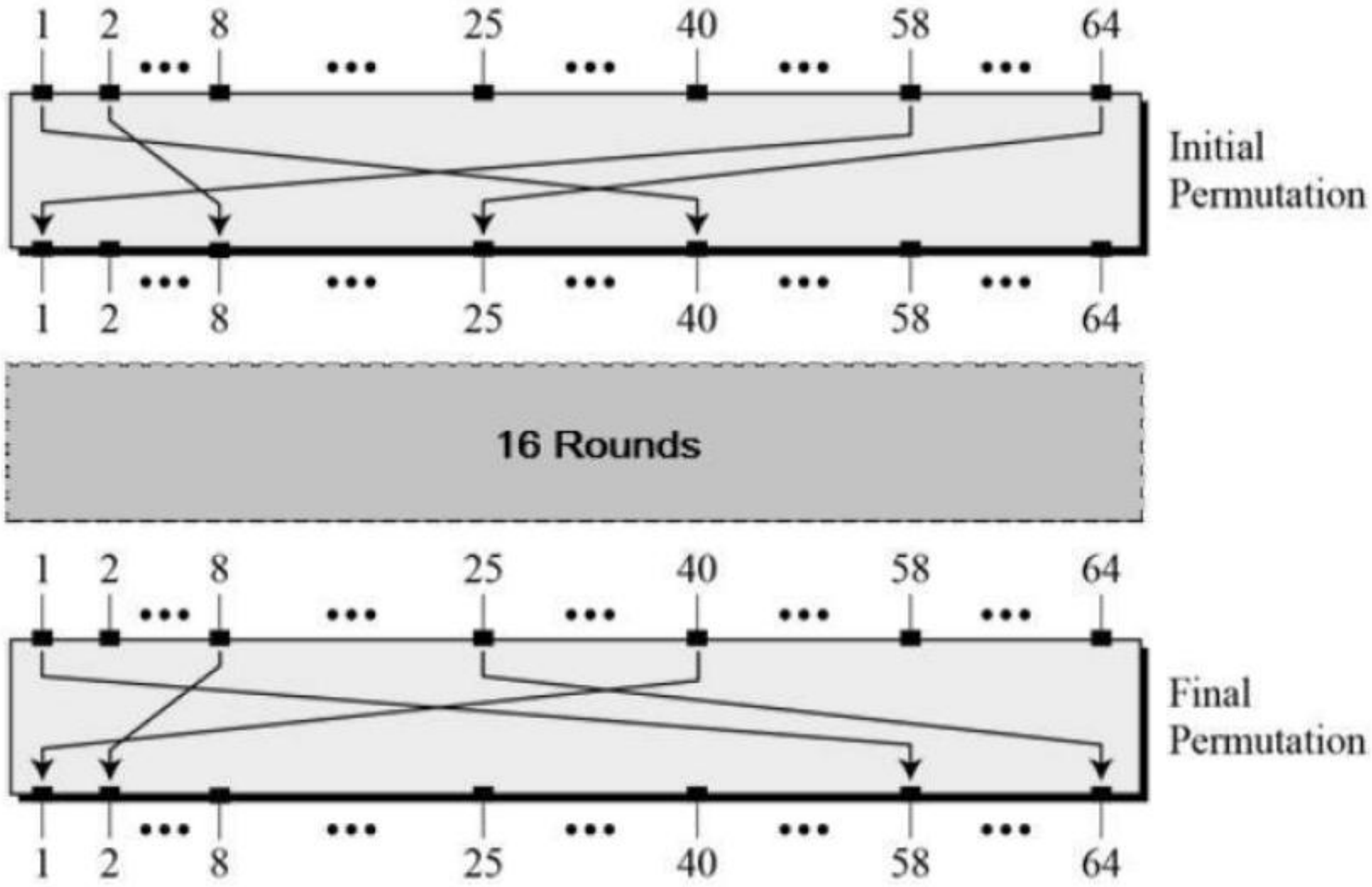
- **Step 1 (Block Division):**
  - First, divide the plaintext into 64-bit blocks (the block size of DES).
- **Step 2 (Padding):**
  - If necessary, pad the final block to make it 64 bits.
- **Step 3: CBC Mode Application:**
  - In CBC mode, the encryption of each block is dependent on the previous ciphertext block.
  - The first block is XORed with an initialization vector (IV).
  - The subsequent blocks are XORed with the previous ciphertext block before being encrypted by DES.
- **Step 4: (Encryption Using DES):**
  - Apply DES (which uses the Feistel structure) to each block individually.
  - DES will use its Feistel network to process each 64-bit block.

General Structure of DES

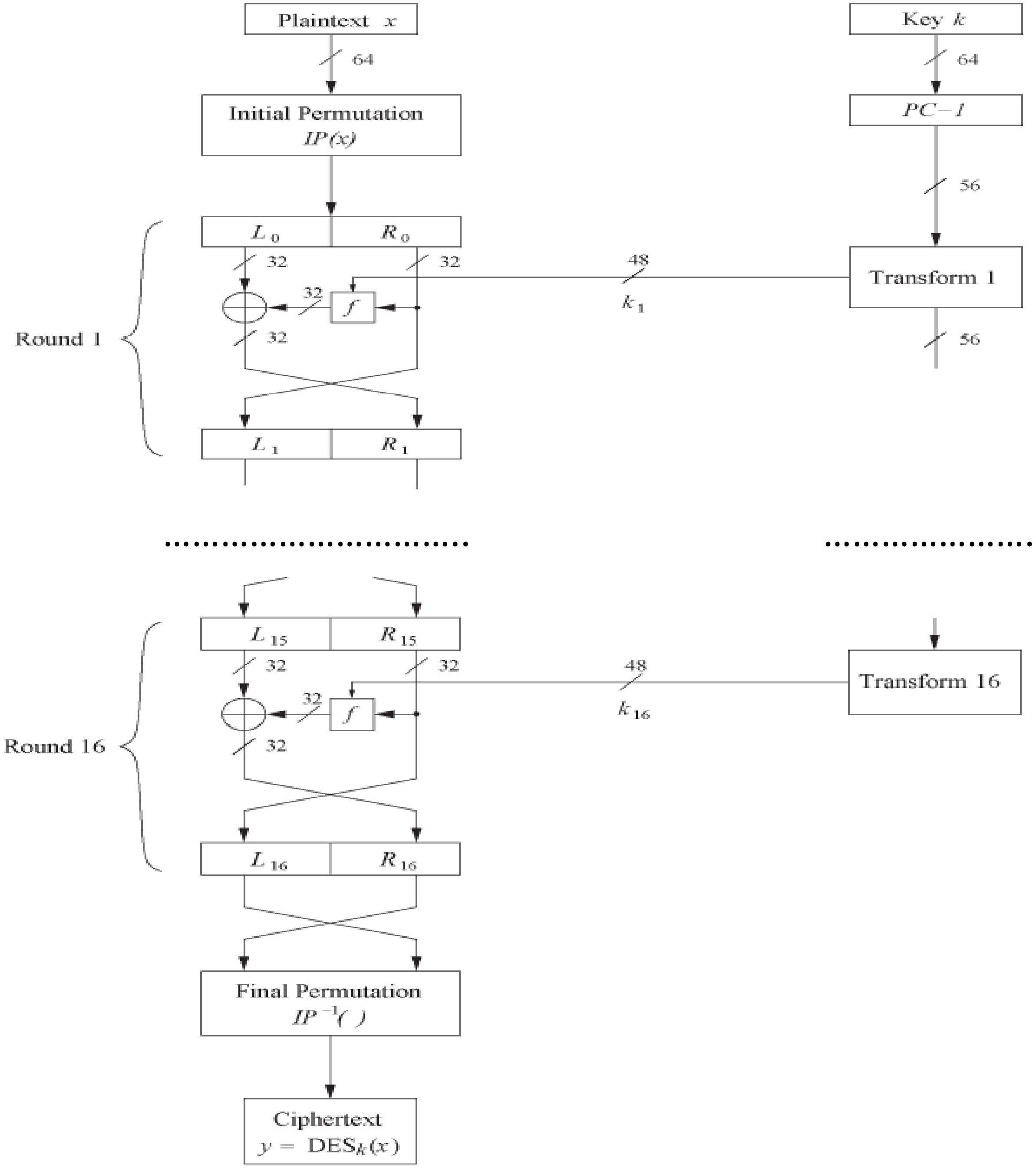
- Major characteristics:
  - Round function
  - Key schedule
  - Initial and final permutation



Initial and Final Permutation



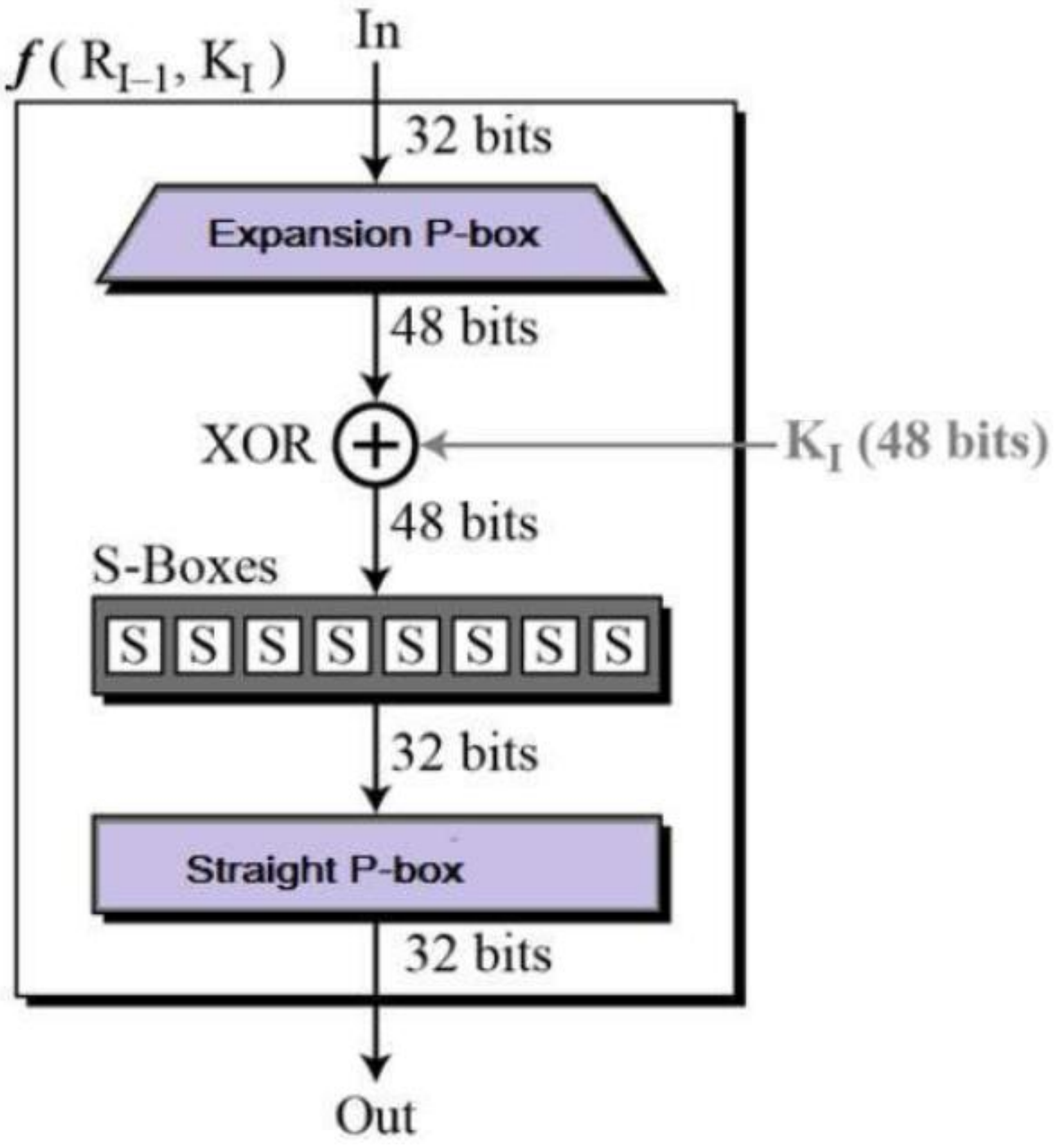
- Permutation (or Transposition):
  - This operation changes the order of bits according to a predefined pattern.
  - No bits are changed in value; only their positions are shuffled.
- The initial and final permutations are straight Permutation boxes (P-boxes) that are inverses of each other.

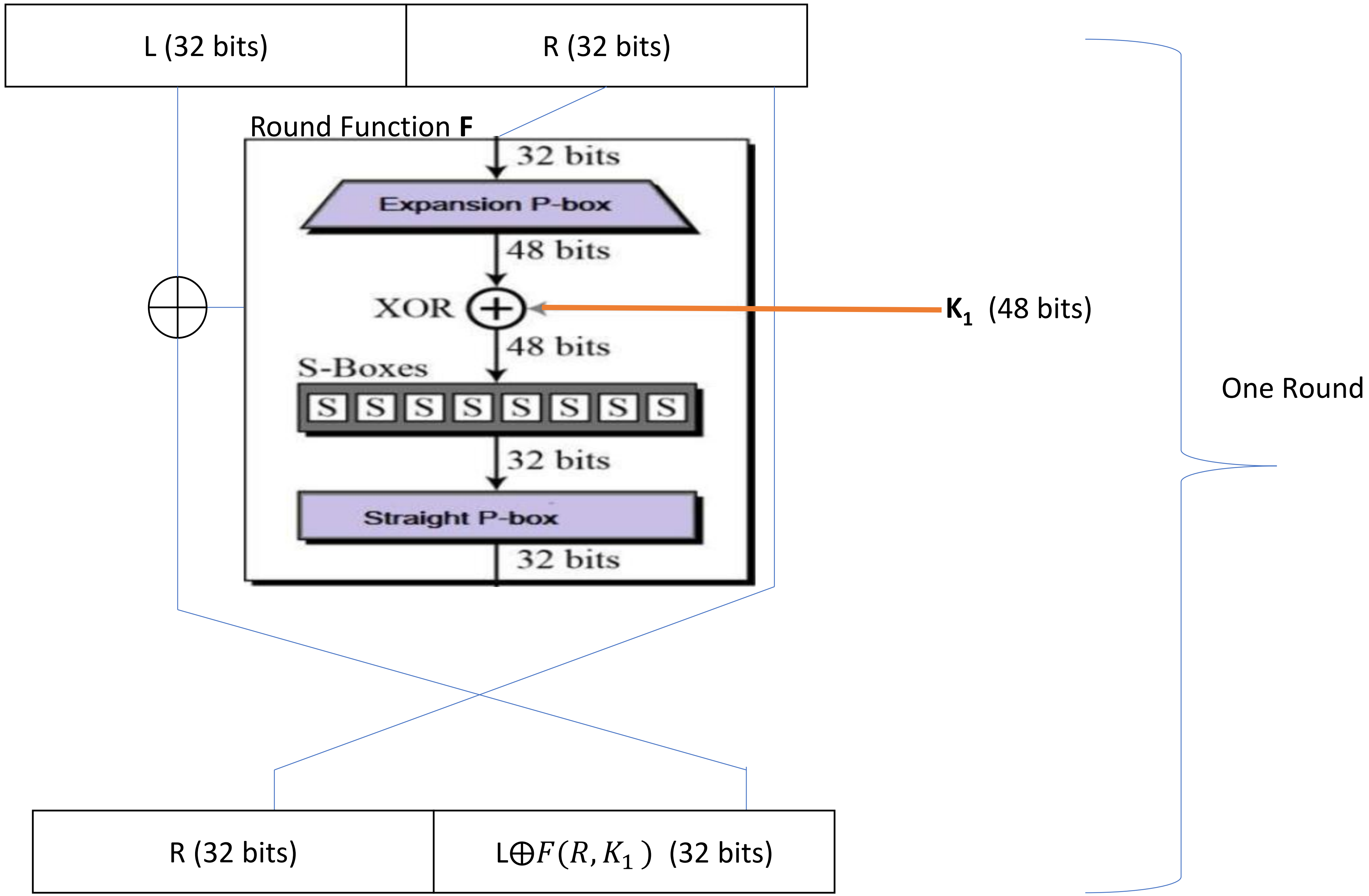


DES Structure

Round Function

- The heart of this cipher is the DES function, f.
- The DES function applies a 48-bit key to the rightmost 32 bits to produce a 32-bit output.



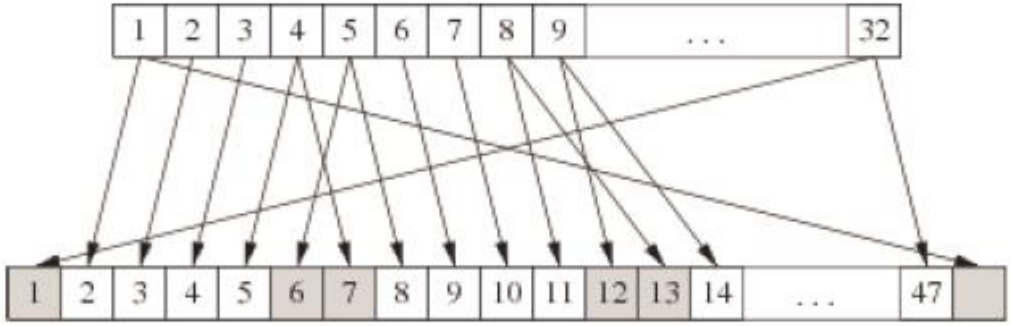

L (32 bits)
R (32 bits)
Round Function F
32 bits
Expansion P-box
48 bits
XOR
K_1 (48 bits)
48 bits
S-Boxes
S S S S S S S S
32 bits
Straight P-box
32 bits
One Round
R (32 bits)
$L \oplus F(R, K_1)$ (32 bits)

Expansion Permutation Box : From 32 to 48 bits by bit repetition

- Expansion Permutation Box –
  - Since right input is 32-bit and round key is a 48-bit, we first need to expand right input to 48 bits.
  - This step ensures that say if there is one bit flip in the bit position 1,32 etc., there will be 2-bit flips in the output of the P-Box.

32 bit original data

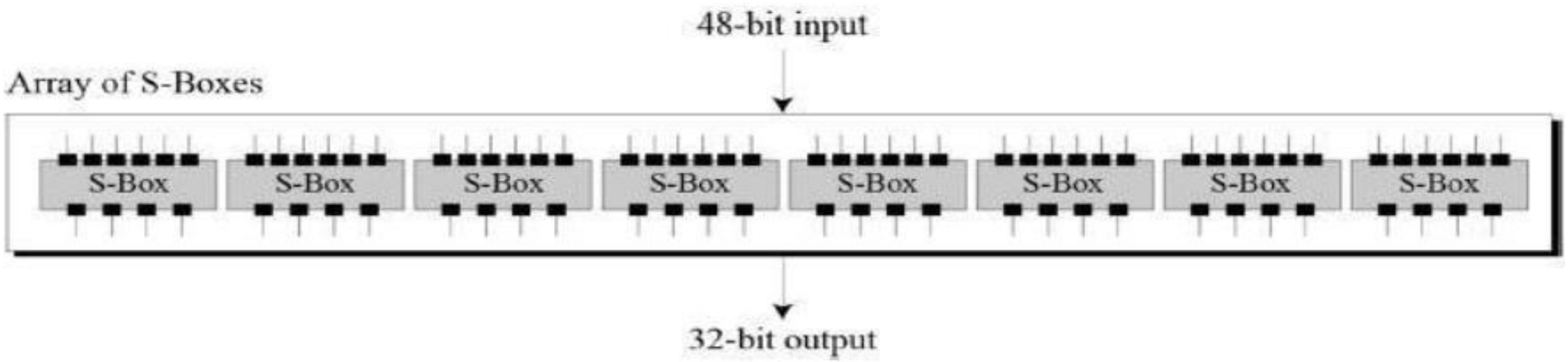
01	02	03	04
05	06	07	08
09	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	31	31	32



Diffusion!!

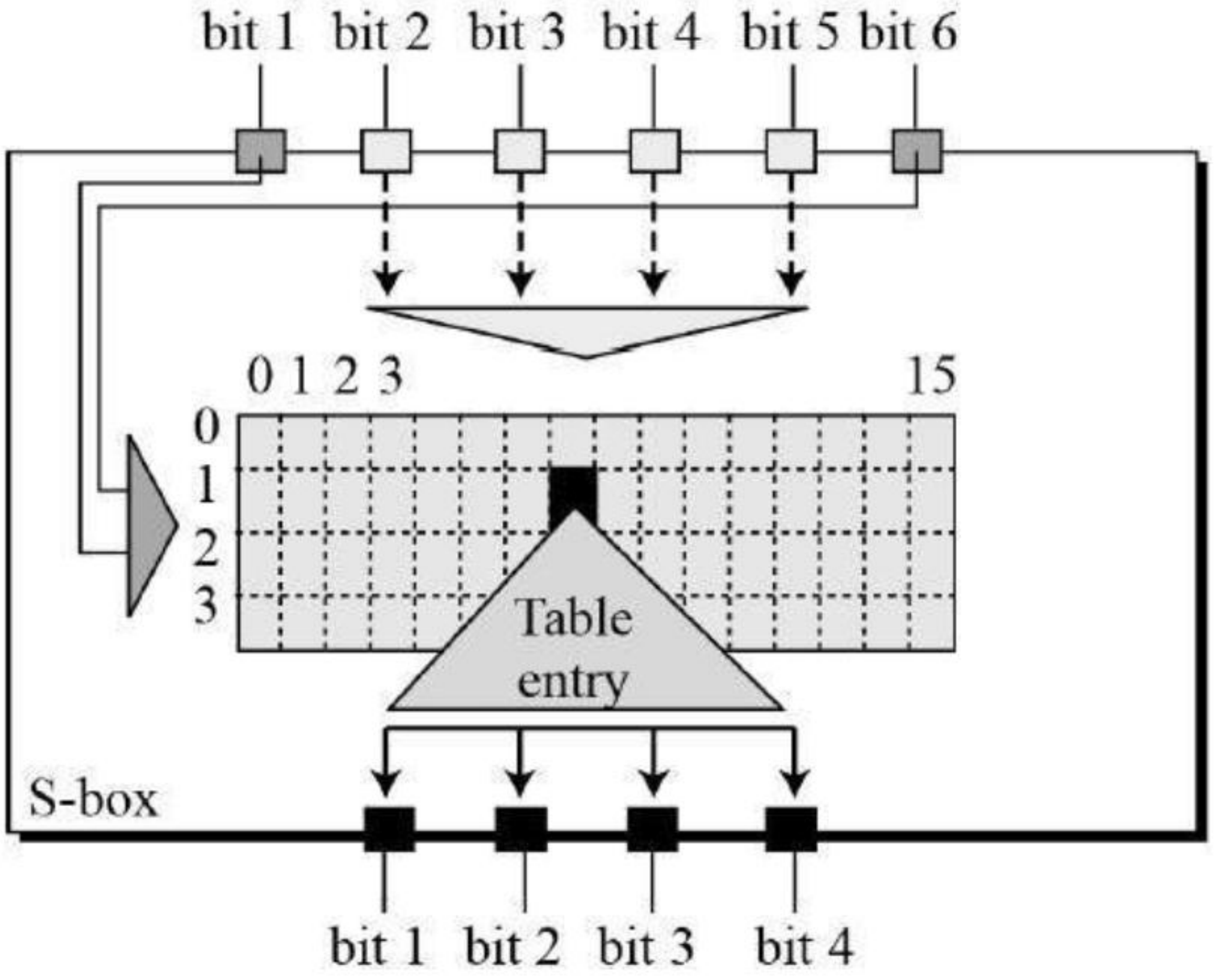
Round Function (cont..)

- XOR –
  - After the expansion permutation, DES does XOR operation on the expanded right section and the round key.
  - The round key is used only in this operation.
- Substitution Boxes. –
  - The S-boxes carry out the real mixing (**confusion**).
  - DES uses 8 S-boxes, each with a 6-bit input and a 4-bit output.
  - Refer the following illustration –



S-Boxes in Round Function

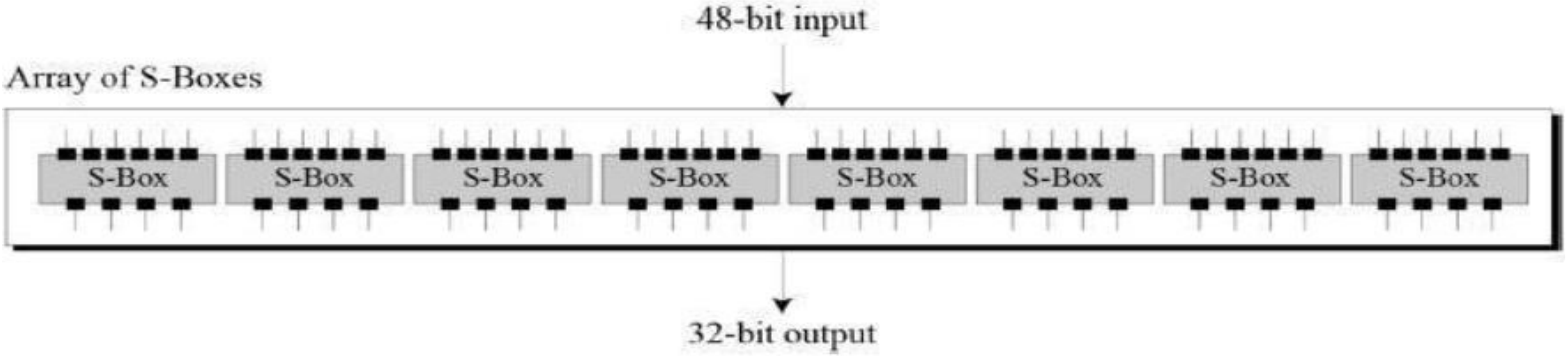
- There are a total of eight S-box tables.
- The output of all eight s-boxes is then combined into 32 bit section.



S-Box example

If the first 6 bits of the 48 bit is **011011**

S_5		Middle 4 bits of input															
		0000	**0001**	**0010**	**0011**	**0100**	**0101**	**0110**	**0111**	**1000**	**1001**	**1010**	**1011**	**1100**	**1101**	**1110**	**1111**
Outer bits	**00**	0010	1100	0100	0001	0111	1010	1011	0110	1000	0101	0011	1111	1101	0000	1110	1001
	01	1110	1011	0010	1100	0100	0111	1101	0001	0101	0000	1111	1010	0011	1001	1000	0110
	10	0100	0010	0001	1011	1010	1101	0111	1000	1111	1001	1100	0101	0110	0011	0000	1110
	11	1011	1000	1100	0111	0001	1110	0010	1101	0110	1111	0000	1001	1010	0100	0101	0011



8 different S-Boxes

Table 3.3 Definition of DES S-Boxes

S_1	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S_2	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S_3	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S_4	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S_5	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S_6	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S_7	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S_8	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Straight Permutation

- Straight Permutation –
- The 32 bit output of S-boxes is then subjected to the straight permutation with rule shown in the following illustration:

16	07	20	21	29	12	28	17
01	15	23	26	05	18	31	10
02	08	24	14	32	27	03	09
19	13	30	06	22	11	04	25

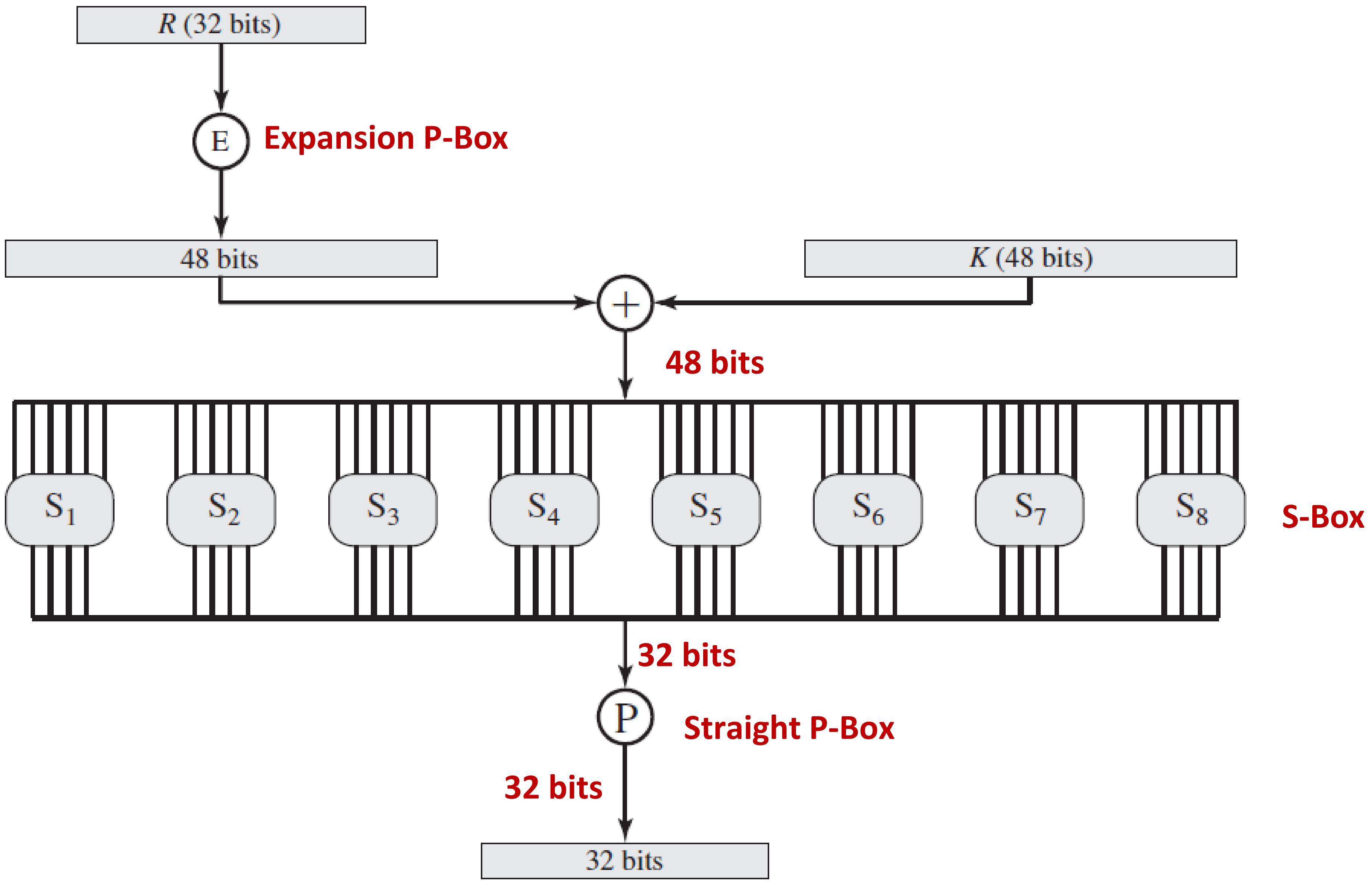
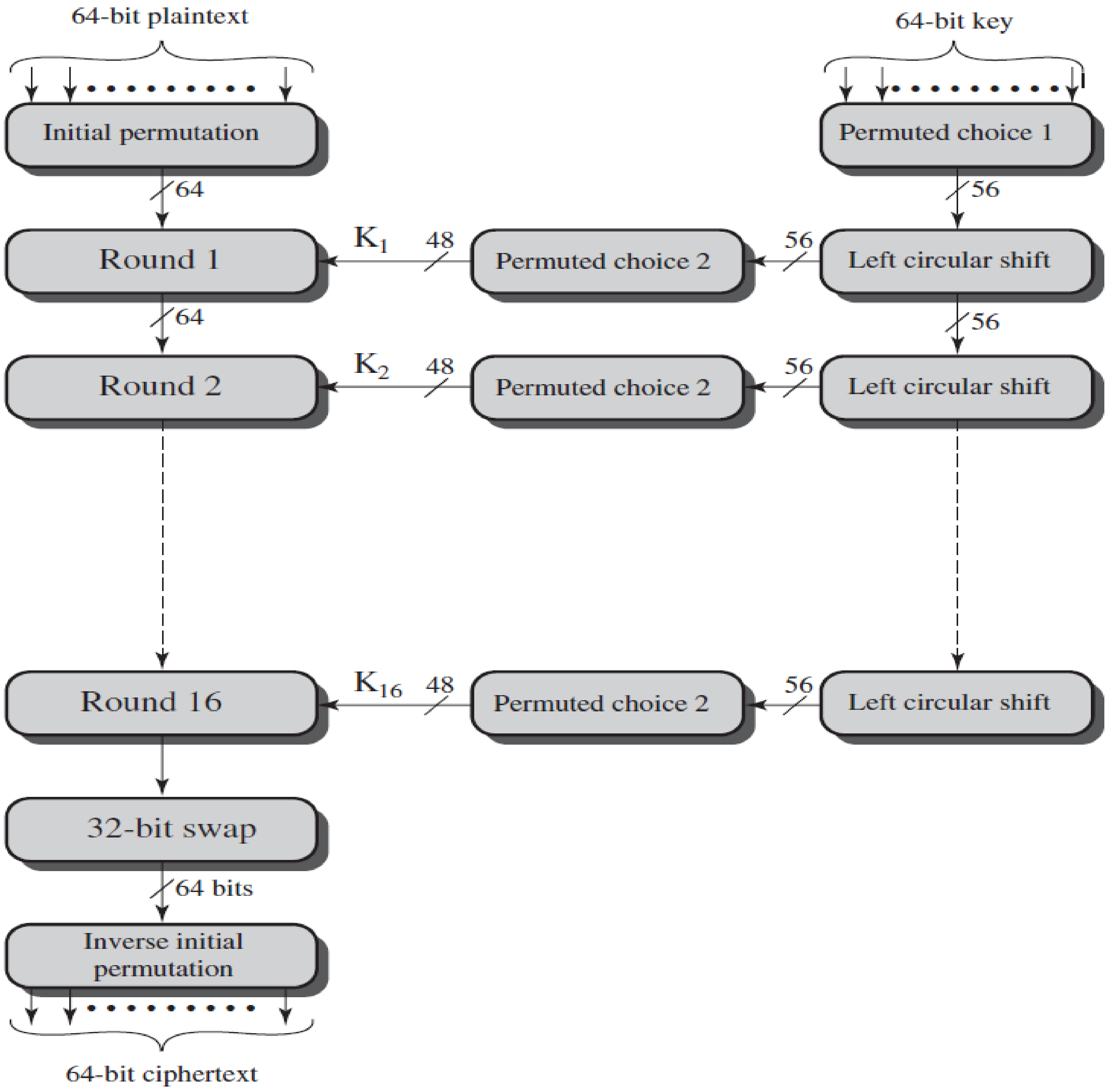


Figure 3.7 Calculation of F(R, K)

DES Architecture



Key Generation: 64 -56 bit conversion

- Let the following be the initial 64 bit key (8 blocks of 8 bits each).
  - 11001100 00111111 00000011 ……………………. 10000010
- Suppose we use odd parity (adjust the 8th bit for parity):
  - 1100110**1** 0011111**0** 0000001**0** ……………………. 1000001**1**
- This process is only for error checking.
- The eighth bit in each block will be discarded for encryption/ decryption.
- So effectively, 56 bits is used for round key generation.
  - 1100110 0011111 0000001 ……………………. 1000001

Key Generation

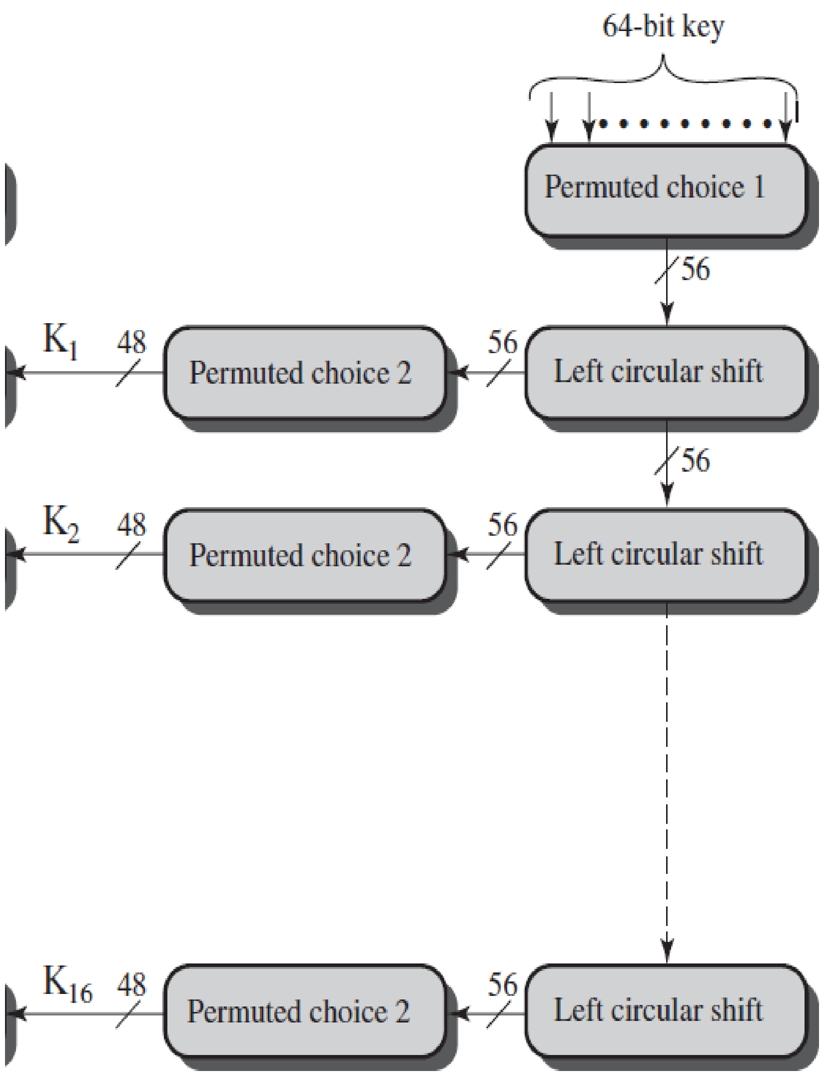
(a) Input Key

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

- A 64-bit key is used as input to the algorithm.
- The bits of the key are numbered from 1 through 64;
- Every eighth bit is ignored.

(b) Permuted Choice One (PC-1)

57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4



- The key is first subjected to a permutation governed by a table labeled Permuted Choice One

Key Generation

- The resulting 56-bit key is then treated as two 28-bit quantities and labeled C_{i-1}, D_{i-1}.
- At each round, and are separately subjected to a **circular left shift** or (rotation) of 1 or 2 bits (as per table d).
- These shifted values serve as input to the next round.
- They also serve as input to the part labeled Permuted Choice Two (Table c), which produces a 48-bit output.

Compression P-Box
56 → 48-bit mapping

(c) Permuted Choice Two (PC-2)

14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

(d) Schedule of Left Shifts

Round Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bits Rotated	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

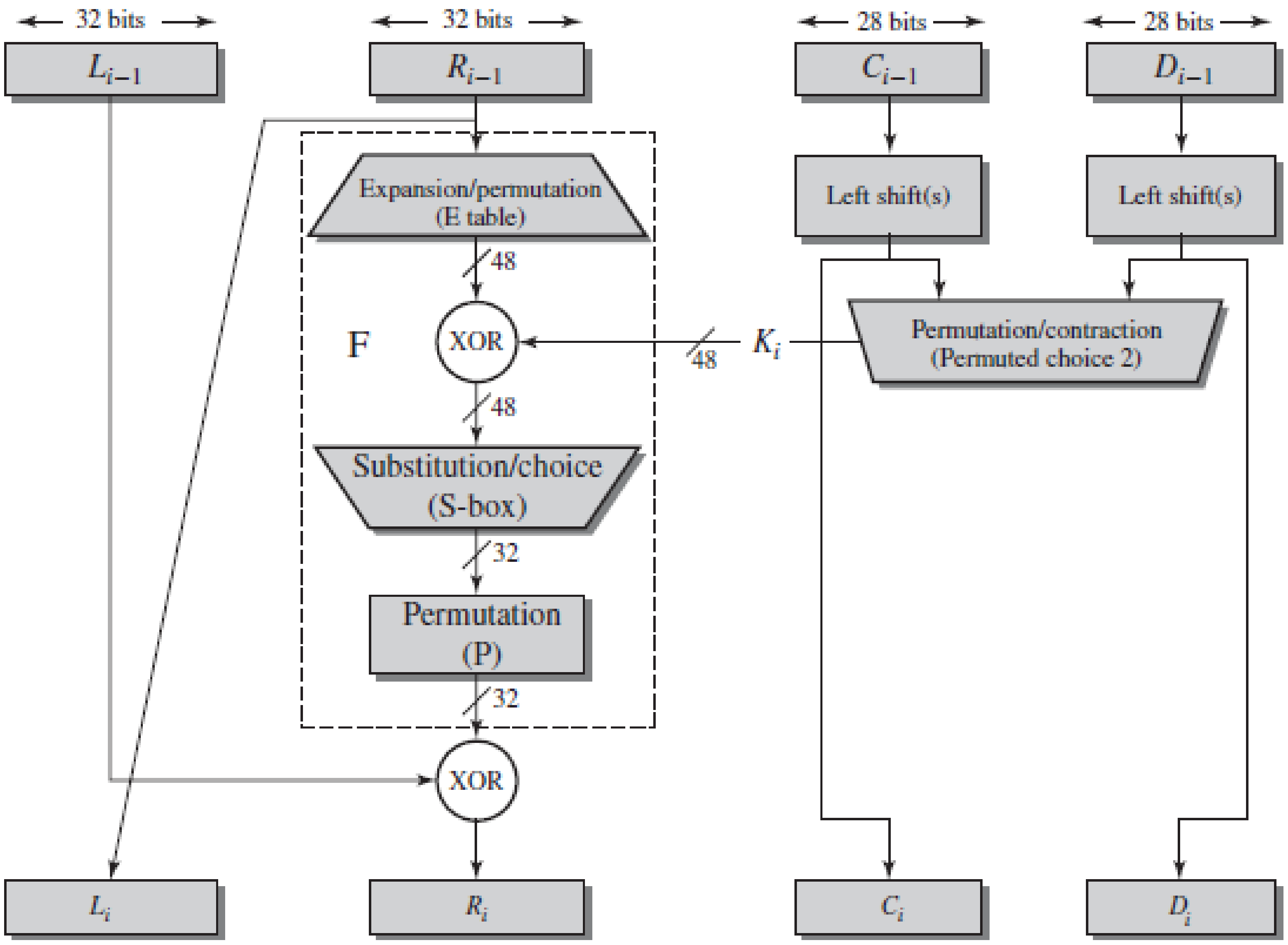
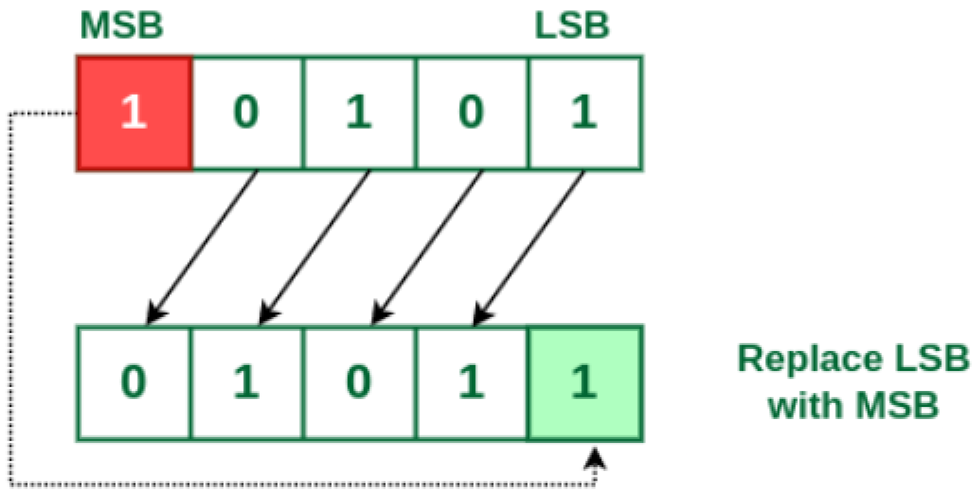


Figure 3.6 Single Round of DES Algorithm

Circular Shift left Operation



Circular Left Shift

- In this shift operation each bit in the register is shifted to the left one by one.
- After shifting, the LSB becomes empty, so the value of the MSB is filled in there.

DES Decryption

- As with any Feistel cipher, decryption uses the same algorithm as encryption, except that the application of the subkeys is reversed.
- **DES Encryption:**
  - **Initial Permutation (IP)**: The plaintext undergoes an initial permutation (IP).
  - **Feistel Rounds**: The permuted plaintext is processed through 16 Feistel rounds using different round keys.
  - **Inverse Initial Permutation (IP^{-1})**: After the rounds, the result is permuted again using the inverse of the initial permutation (IP^{-1}) to produce the final ciphertext.
- **DES Decryption:**
  - **Initial Permutation (IP)**: The ciphertext is subjected to the same initial permutation (IP) as in encryption.
  - **Feistel Rounds** (with reversed key schedule): The permuted ciphertext undergoes the same 16 Feistel rounds but with the round keys applied in reverse order.
  - **Inverse Initial Permutation (IP^{-1})**: Finally, the result is permuted using the inverse initial permutation (IP^{-1}) to produce the original plaintext.

Avalanche effect in DES

- A desirable property in any encryption algorithm.
- Avalanche effect:
  - a small change in either the plaintext or the key should produce a significant change in the ciphertext.
- In DES:
  - 1 bit change in the PT – 34-bits change in CT on an average.
  - 1 bit change in Key – 25-bit change in the CT on an average.

Avalanche Effect: Change Ciphertext

Plaintext:	02468aceeca86420
Key:	0f1571c947d9e859
Ciphertext:	da02ce3a89ecac3b

Table 3.5 DES Example

Round	K_i	L_i	R_i
IP		5a005a00	3cf03c0f
1	1e030f03080d2930	3cf03c0f	bad22845
2	0a31293432242318	bad22845	99e9b723
3	23072318201d0c1d	99e9b723	0bae3b9e
4	05261d3824311a20	0bae3b9e	42415649
5	3325340136002c25	42415649	18b3fa41
6	123a2d0d04262a1c	18b3fa41	9616fe23
7	021f120b1c130611	9616fe23	67117cf2
8	1c10372a2832002b	67117cf2	c11bfc09
9	04292a380c341f03	c11bfc09	887fbc6c
10	2703212607280403	887fbc6c	600f7e8b
11	2826390c31261504	600f7e8b	f596506e
12	12071c241a0a0f08	f596506e	738538b8
13	300935393c0d100b	738538b8	c6a62c4e
14	311e09231321182a	c6a62c4e	56b0bd75
15	283d3e0227072528	56b0bd75	75e8fd8f
16	2921080b13143025	75e8fd8f	25896490
IP^{-1}		da02ce3a	89ecac3b

Note: DES subkeys are shown as eight 6-bit values in hex format

Plaintext:	02468aceeca86420
Key:	0f1571c947d9e859
Ciphertext:	da02ce3a89ecac3b

Plaintext:	12468aceeca86420
Key:	0f1571c947d9e859
Ciphertext:	057cde97d7683f2a

Round		δ
	02468aceeca86420 12468aceeca86420	1
1	3cf03c0fbad22845 3cf03c0fbad32845	1
2	bad2284599e9b723 bad3284539a9b7a3	5
3	99e9b7230bae3b9e 39a9b7a3171cb8b3	18
4	0bae3b9e42415649 171cb8b3ccaca55e	34
5	4241564918b3fa41 ccaca55ed16c3653	37
6	18b3fa419616fe23 d16c3653cf402c68	33
7	9616fe2367117cf2 cf402c682b2cefbc	32
8	67117cf2c11bfc09 2b2cefbc99f91153	33

Round		δ
9	c11bfc09887fbc6c 99f911532eed7d94	32
10	887fbc6c600f7e8b 2eed7d94d0f23094	34
11	600f7e8bf596506e d0f23094455da9c4	37
12	f596506e738538b8 455da9c47f6e3cf3	31
13	738538b8c6a62c4e 7f6e3cf34bc1a8d9	29
14	c6a62c4e56b0bd75 4bc1a8d91e07d409	33
15	56b0bd7575e8fd8f 1e07d4091ce2e6dc	31
16	75e8fd8f25896490 1ce2e6dc365e5f59	32
IP^{-1}	da02ce3a89ecac3b 057cde97d7683f2a	32

Avalanche Effect: Change in Key

Plaintext:	02468aceeca86420
Key:	0f1571c947d9e859
Ciphertext:	da02ce3a89ecac3b

Plaintext:	02468aceeca86420
Key:	1f1571c947d9e859
Ciphertext:	ee92b50606b62b0b

Table 3.7 Avalanche Effect in DES: Change in Key

Round		δ
	02468aceeca86420 02468aceeca86420	0
1	3cf03c0fbad22845 3cf03c0f9ad628c5	3
2	bad2284599e9b723 9ad628c59939136b	11
3	99e9b7230bae3b9e 9939136b768067b7	25
4	0bae3b9e42415649 768067b75a8807c5	29
5	4241564918b3fa41 5a8807c5488dbe94	26
6	18b3fa419616fe23 488dbe94aba7fe53	26
7	9616fe2367117cf2 aba7fe53177d21e4	27
8	67117cf2c11bfc09 177d21e4548f1de4	32

Round		δ
9	c11bfc09887fbc6c 548f1de471f64dfd	34
10	887fbc6c600f7e8b 71f64dfd4279876c	36
11	600f7e8bf596506e 4279876c399fdc0d	32
12	f596506e738538b8 399fdc0d6d208dbb	28
13	738538b8c6a62c4e 6d208dbbb9bdeeaa	33
14	c6a62c4e56b0bd75 b9bdeeaad2c3a56f	30
15	56b0bd7575e8fd8f d2c3a56f2765c1fb	33
16	75e8fd8f25896490 2765c1fb01263dc4	30
IP^{-1}	da02ce3a89ecac3b ee92b50606b62b0b	30

The strength of DES

- The use of 56 bit keys (Major weakness of the algorithm).
- Bruteforce - 2^{56} possible keys $\approx 7.2 \times 10^{16}$ keys
- A bruteforce appears impractical.
- A machine that can do one DES encryption/decryption per microsecond takes more than thousand years to break the cipher.
- Suppose we have a cluster of such machines and we divide the keyspace among them for bruteforce.
  - If you had 1 million machines, you could divide the keyspace into 1 million parts, with each machine responsible for checking $\frac{2^{56}}{1000{,}000}$ keys.
    - 10 hours to break.
  - DES was finally broken in 1998 (DES cracker, Electronic Frontier Foundation (EFF))
    - The DES cracker divided the keyspace among the **different chips** and **processed them in parallel**.
    - Tested around **90 billion keys per second**, which enabled it to find the correct key in **less than three days**.
    - Cost = 250,000 dollars in 1998.